



"PPEC" REGENERATIVE SERIES FILTER

INSTALLATION MANUAL

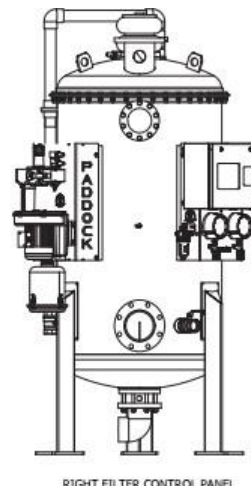
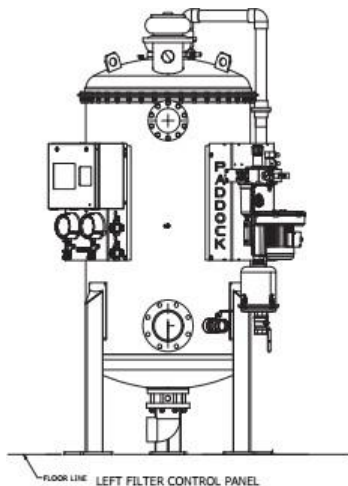




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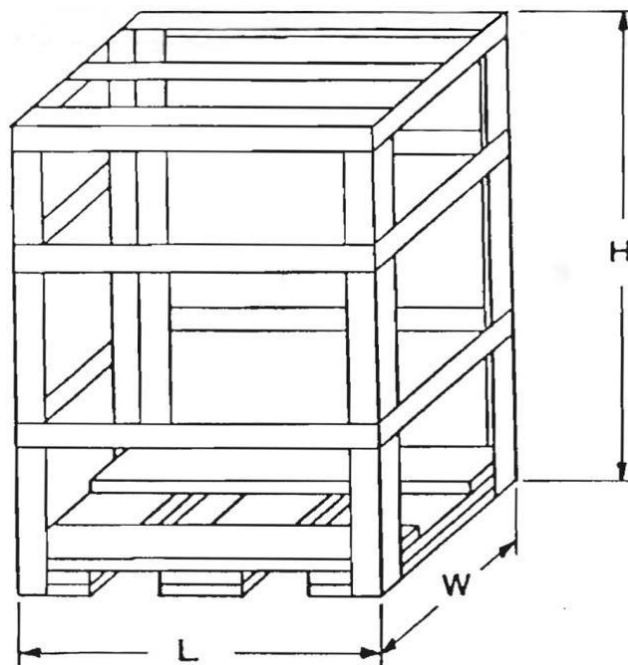
INSTALLATION MANUAL

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Shipping Data

PPEC Regenerative filters are skidded for forklift handling. Filters must be transported and handled only in a **vertical position** or serious internal damage can occur.

Lifting lugs are provided on heads of PPEC 350S and larger models.

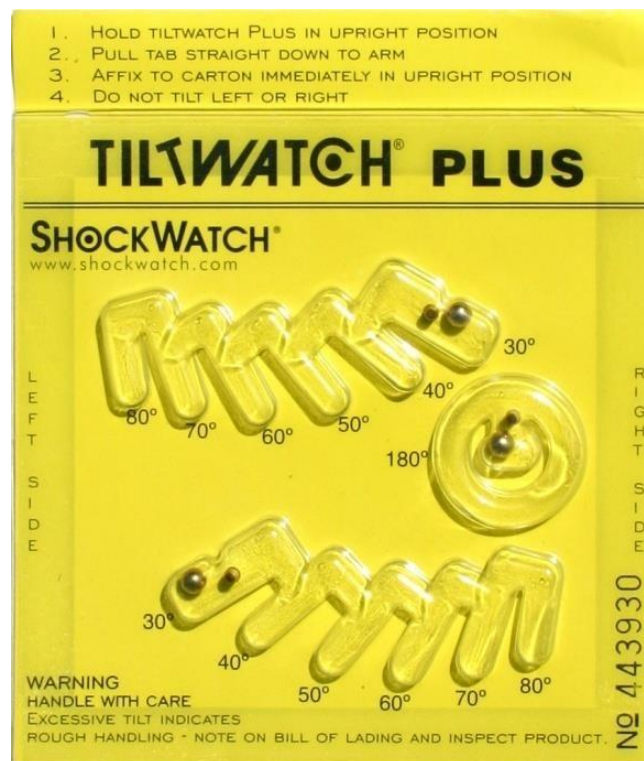


FILTER MODEL	W (in)	L (in)	H (in)	SHIPPING WEIGHT (lbs)
PPEC 140	45 ½	45 ½	81 ¼	310
PPEC 145	49 ½	49 ½	77 ¾	604
PPEC 225	49 ½	49 ½	85 ¼	875
PPEC 225S	49 ½	49 ½	93 ¼	1025
PPEC 350	66	53	87 ½	1150
PPEC 350S	66	53	95 ½	1300
PPEC 500	76	60	89 ¾	1530
PPEC 500S	76	60	97 ¾	1750
PPEC 700	79	66 ½	82 ¼	1980
PPEC 700S	79	66 ½	90 ¼	2200
PPEC 900	84 ½	70	87	2530
PPEC 900S	84 ½	70	95	2750
PPEC 1000	86 ½	74 ½	89 1/8	2880
PPEC 1275	86 ½	74 ½	97 7/8	3100
PPEC 1400	90 ½	82	97 1/8	3880
PPEC 1400S	90 ½	82	97 1/8	4100
PPEC 1675	94 ½	86 ½	95 ½	4500
PPEC 2100	94 ½	86 ½	108	5800

Filter Handling

All PPEC series filters contain precision machined internal parts. It is very important the filter is transported and otherwise handled in an upright position or serious internal damage could result.

All PPEC series filters are shipped with a tilt indicator (See example below) which provides a reference for any deviation from the upright position during shipping and installation. If you receive shipment of the filter with the tilt indicators not in the shown positions below, contact Paddock Pool Equipment Company Customer Service before installation.



NOTE:

It is critical that the filter remains upright while being moved into place. If you find the filter must be tilted, stop and contact Paddock Pool Equipment Company first.

Failure to do so will void the filter warranty and could result in costly repairs.

Filter Location and Clearances

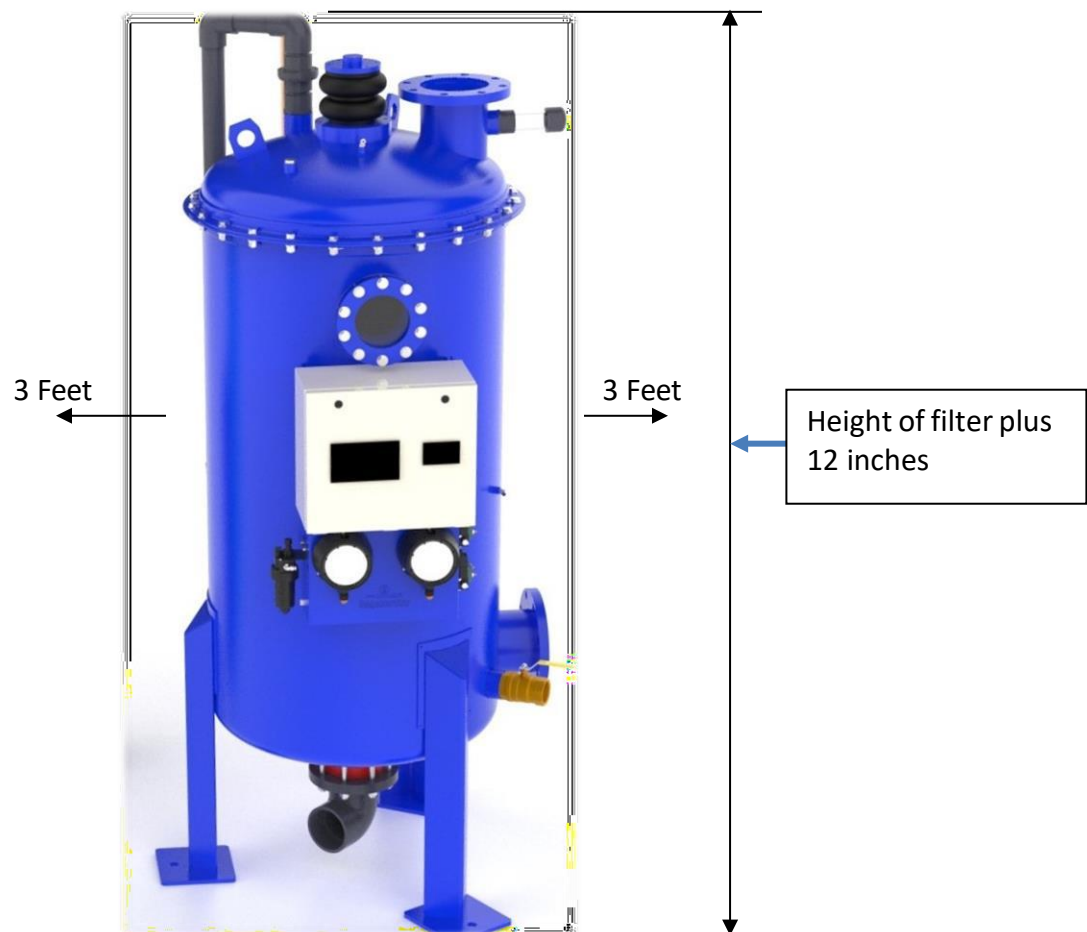
The filter should be housed in a dry ventilated area protected from freezing temperatures. However, if necessary, the filter may be located outdoors provided all electrical and pneumatic components are protected from weather exposure.

Side clearance of at least three feet (3 FT) should be provided around the filter between other components in the system and walls/doorways.

Minimum vertical clearance over the filter is twelve inches (12"), however greater clearance will improve convenience of service.

A suitably anchored eye-bolt or hoist (one-ton minimum safe load) above the filter should be installed to facilitate head removal.

Typical clearance underneath the filter (measured from face of drain flange to floor) is fourteen inches.

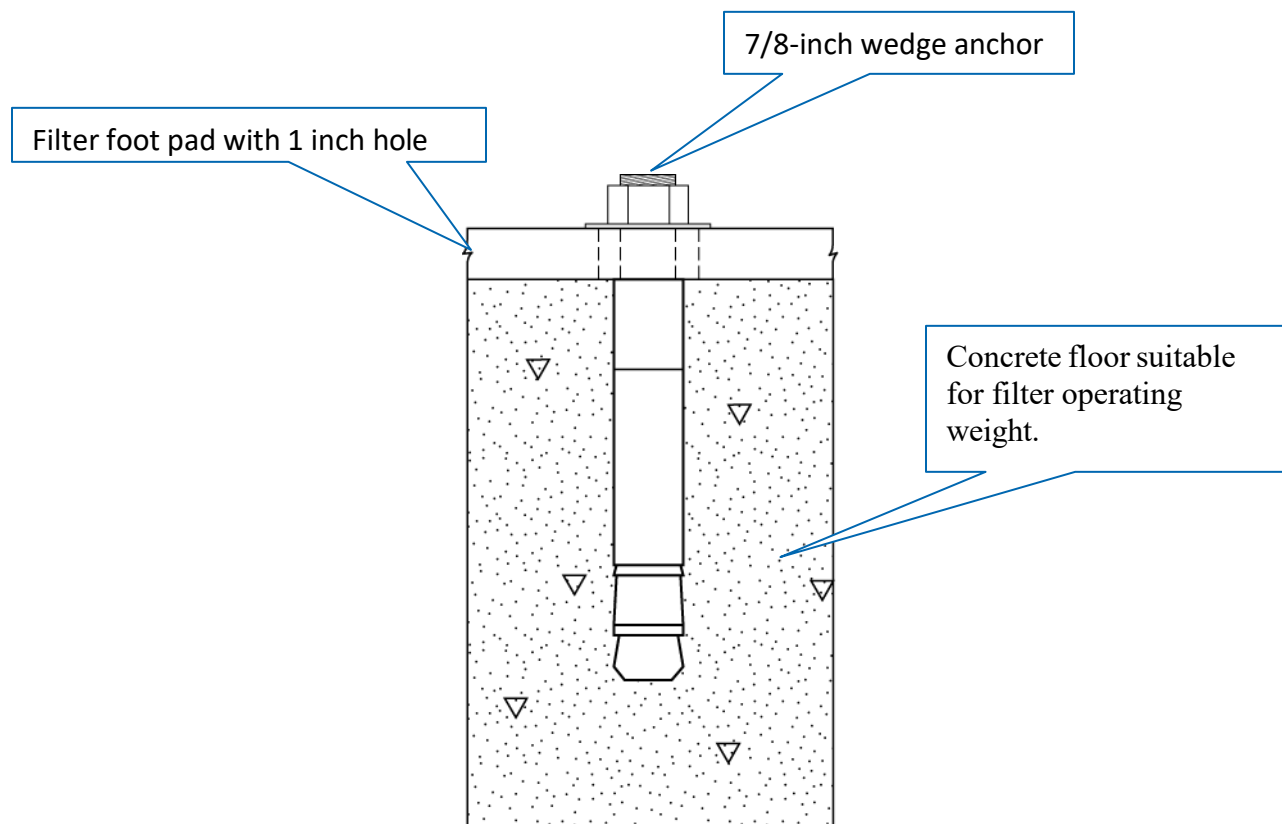


Filter Anchoring Detail

The filter arrives full assembled (unless otherwise noted) and ready to be installed.

Provide a level, reinforced concrete floor or base under the filter. Shim and grout leg pads as needed to make sure the filter is level.

Once in place, the filter should be secured to the pad or floor using concrete wedge anchors. One inch diameter holes for anchoring are provided in the foot pads.



Note:

Use concrete wedge anchors (minimum of 3") to bolt tank to floor. Use anchor 1/8" less dia. size of thru hole. Follow anchor manufacturer recommendation.

Drain Requirements

Gravity drainage of filter is needed to properly waste heavily laden media.

The drainage piping should be sloped away from the filter and terminate in an open sump or sewer connection. (Check local codes for air-break requirements and media discharge containment.)

During the filter dump operation, the drainage rate is controlled at approximately 50 GPM. When possible, the drainage plumbing should be designed for an emptying rate of 500 GPM to provide adequate run-off capacity in case of operator error.

In cases where the sewer is higher than the filter drain, a gravity sump should be provided with a sump pump to lift the waste from the sump to the sewer.

Please see the chart below for the minimum sump size for each filter in the series.

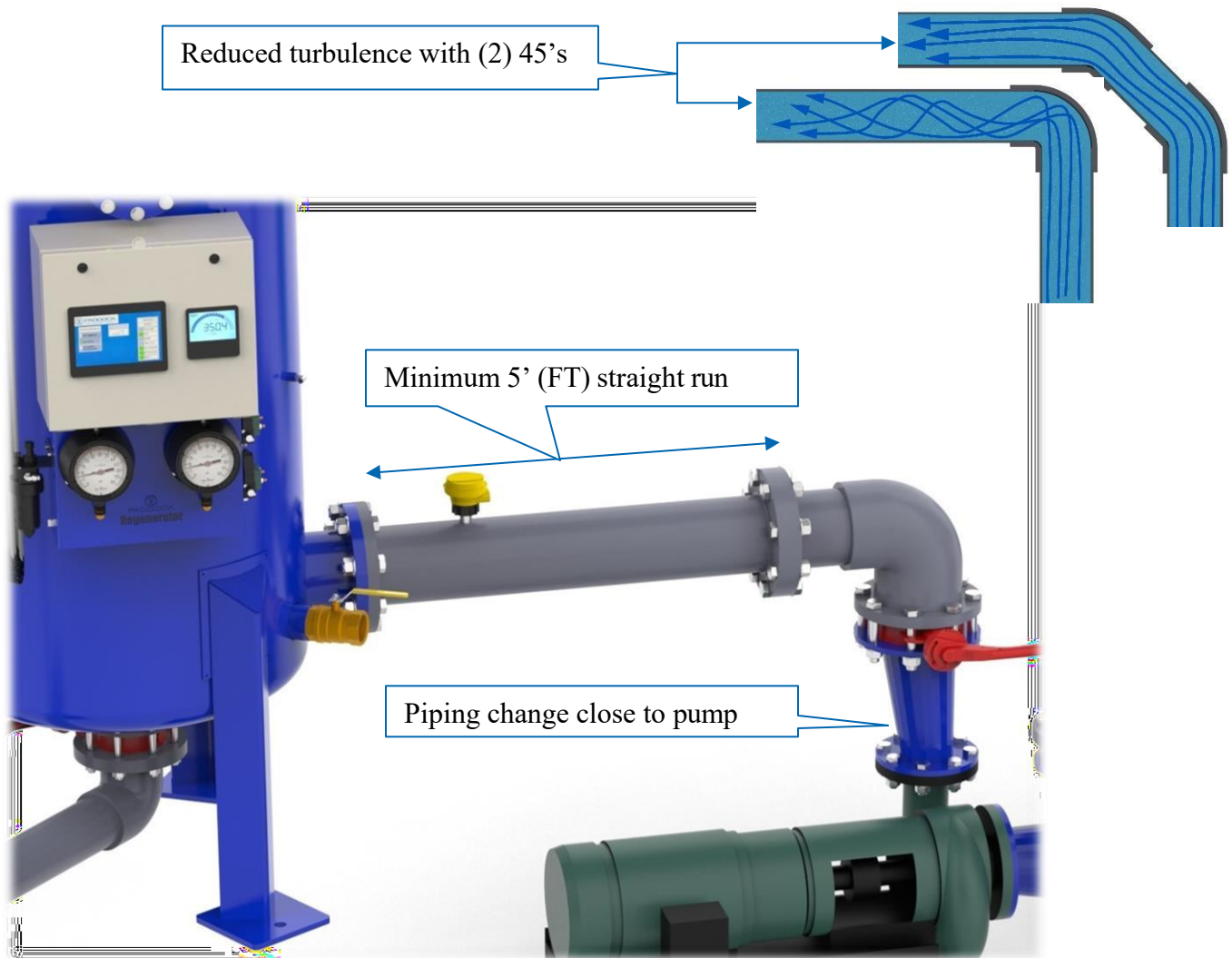
Minimum Sump Size	
PPEC 140	60 gals.
PPEC 145	70 gals
PPEC 225	100 gals.
PPEC 225S	100 gals.
PPEC 350	165 gals.
PPEC 350S	165 gals.
PPEC 500	225 gals.
PPEC 500S	225 gals.
PPEC 700	390 gals.
PPEC 700S	390 gals.
PPEC 900	446 gals.
PPEC 900S	446 gals.
PPEC 1000	670 gals.
PPEC 1275	670 gals.
PPEC 1400	810 gals.
PPEC 1400S	810 gals.
PPEC 1675	960 gals.
PPEC 2100	960 gals.

Plumbing

The PPEC series of filters will operate efficiently above or below pool level. A horizontal check valve should be installed either in front of the pump strainer or between the pump strainer and pre-coat tee. If the pump is located below water level it is suggested a check valve be installed above the pump discharge as well.

Filter Influent Piping

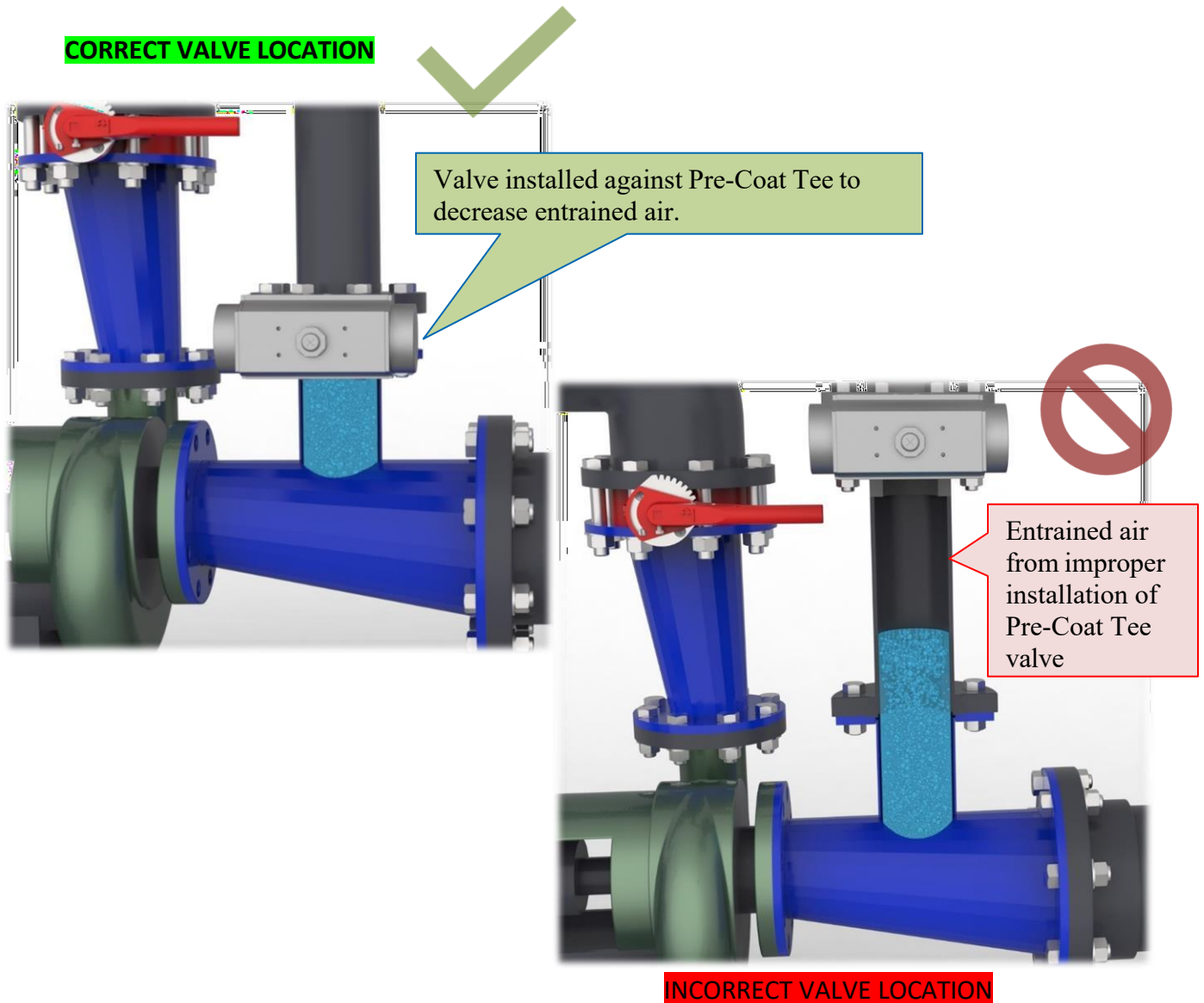
A minimum 5' straight run of pipe going into the influent connection is needed for the filter to operate efficiently; however, proper installation of the flow meter requires 10x the pipe diameter in front of and 5x the diameter behind (see flowmeter location). It is also important to reduce the number of fittings used when connecting the pump discharge piping to the filter. If the 5' (feet) minimum is unable to be met, use two (2) 45's should be used to decrease turbulence. Any increase or reduction in piping size should be done as close to the pump discharge as possible.



Pre-coat Valve Location

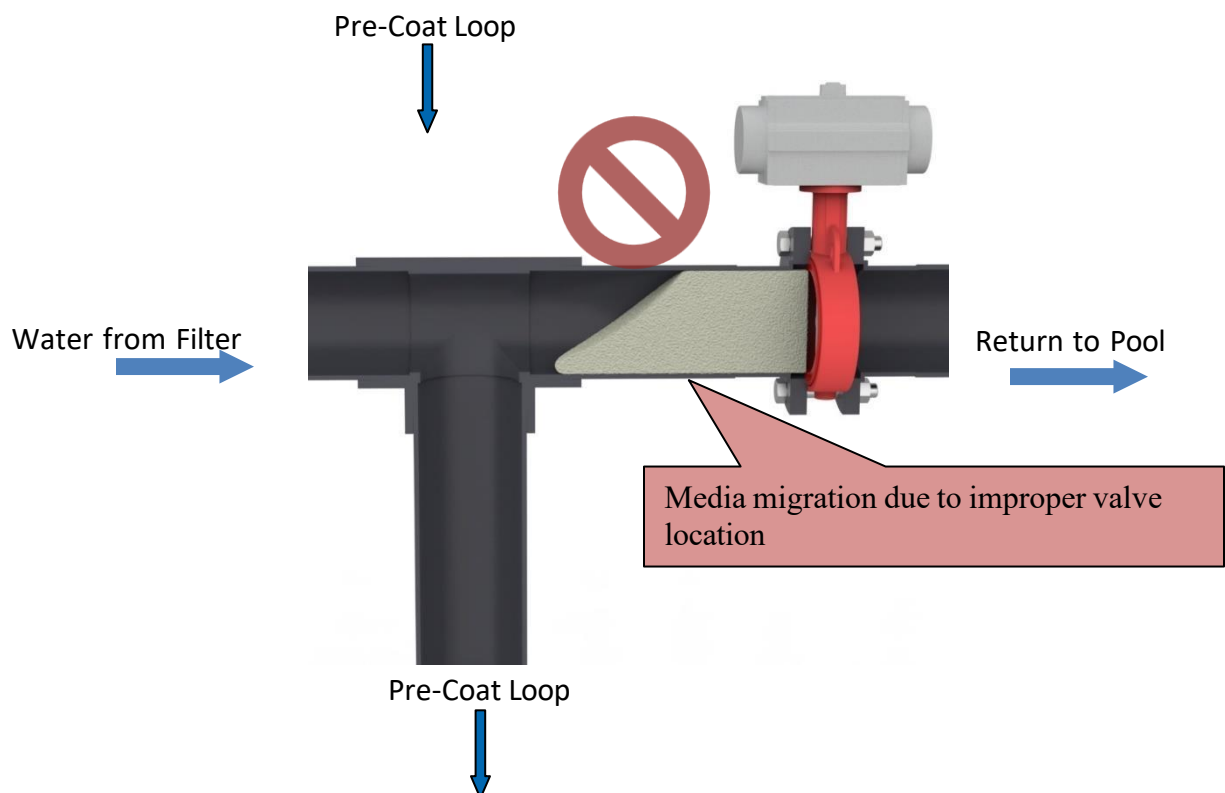
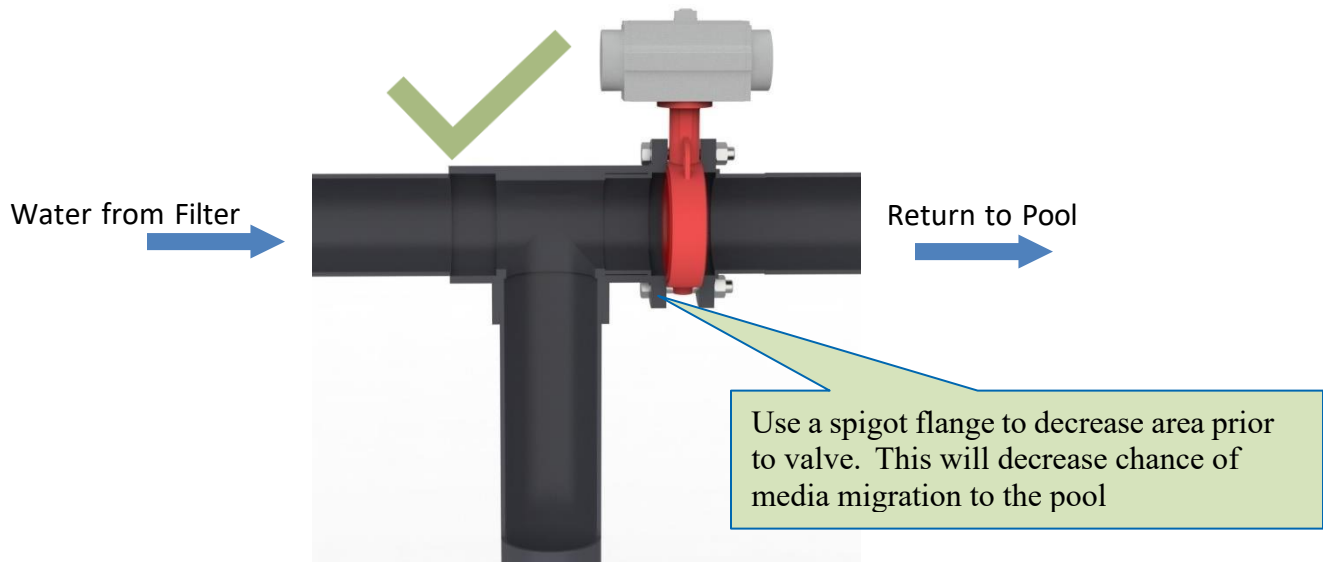
The pre-coat valve needs to be installed as close as possible to the pre-coat tee between the pump and strainer. If this is not done, the resulting entrained air can be pushed into the filter and cause media separation which will result in media getting into the pool.

The pre-coat line and valve are one significant pipe size smaller than the filter connections.



On-stream Valve Location

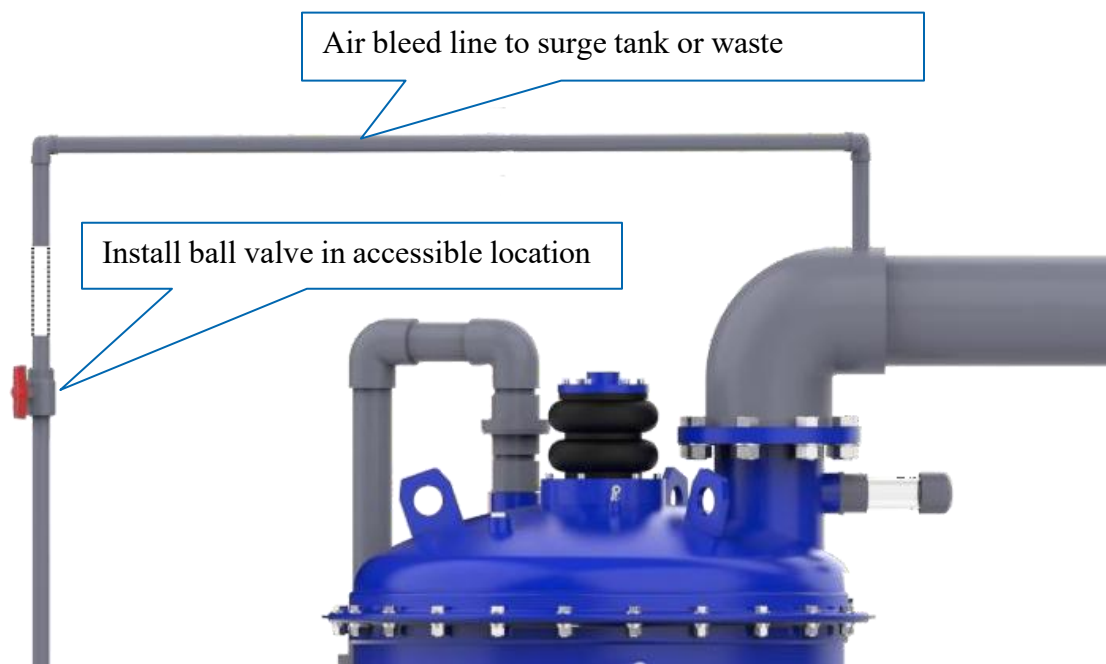
The on-stream valve needs to be installed directly against the on-stream tee. A spigot flange should be used in the tee to decrease the amount of space that media can collect.



High Vent Air Bleed

The air bleed must be installed at the highest point in the return line between the filter effluent connection and the on-stream valve. This line should be 3/4" to 1" in size with a valve located in an accessible place.

If the air bleed is improperly installed it could result in entrained air and decreased filter performance.



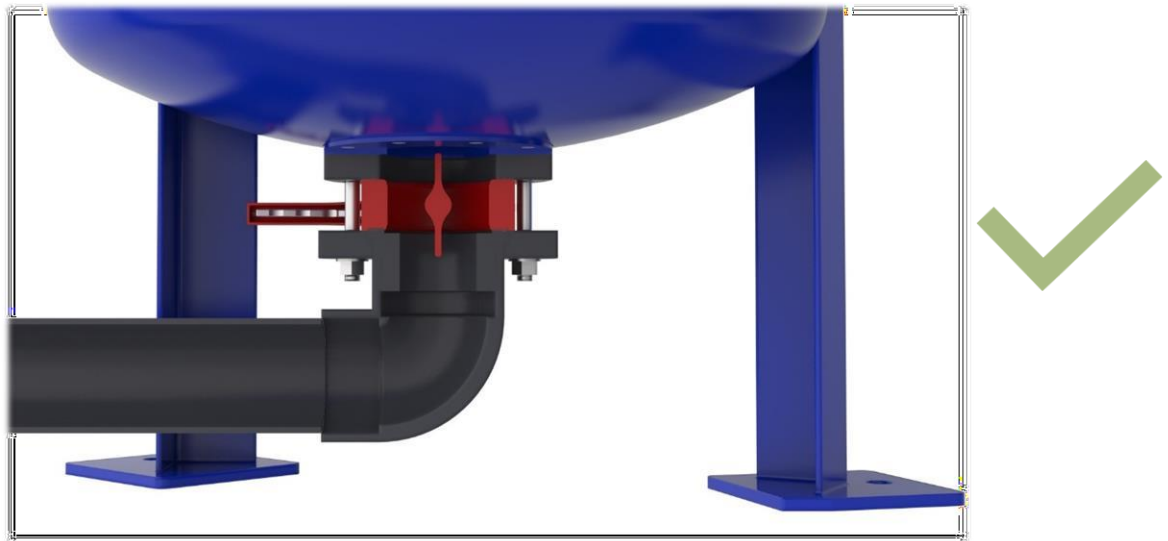
Note:

If the Pre-coat piping is higher than the filter effluent piping, the air bleed line must be installed at the highest point. Failure to do so will not allow air to purge from the piping.

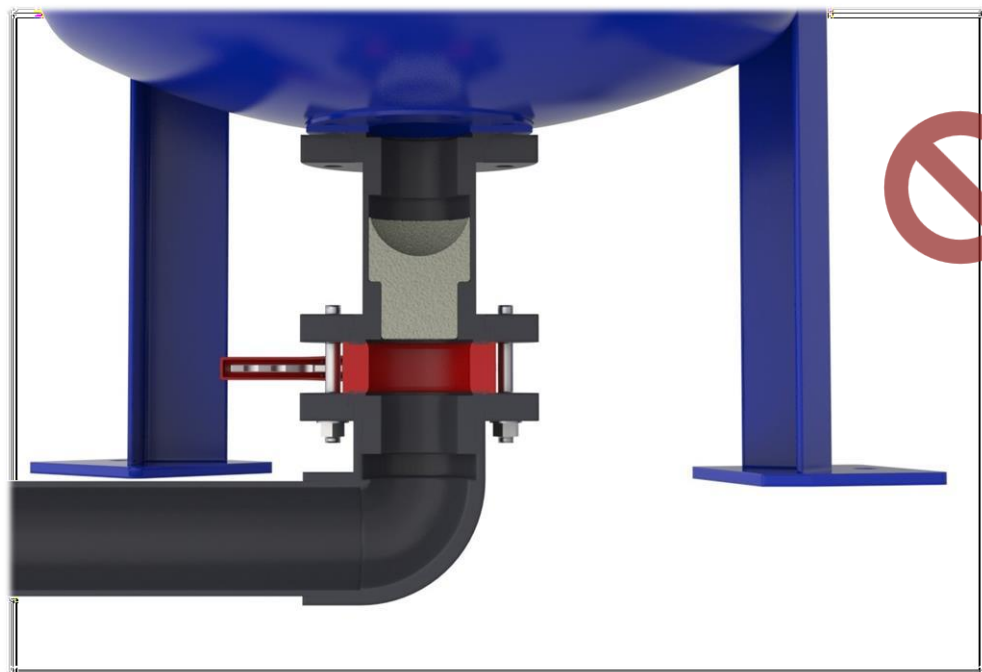
Filter Drain Valve

The filter drain valve needs to be installed directly to the bottom of the filter. If this valve is not installed against the filter, it will cause media to migration into the drain piping causing the filter charge to become light and make it difficult to open the valve.

CORRECT VALVE LOCATION

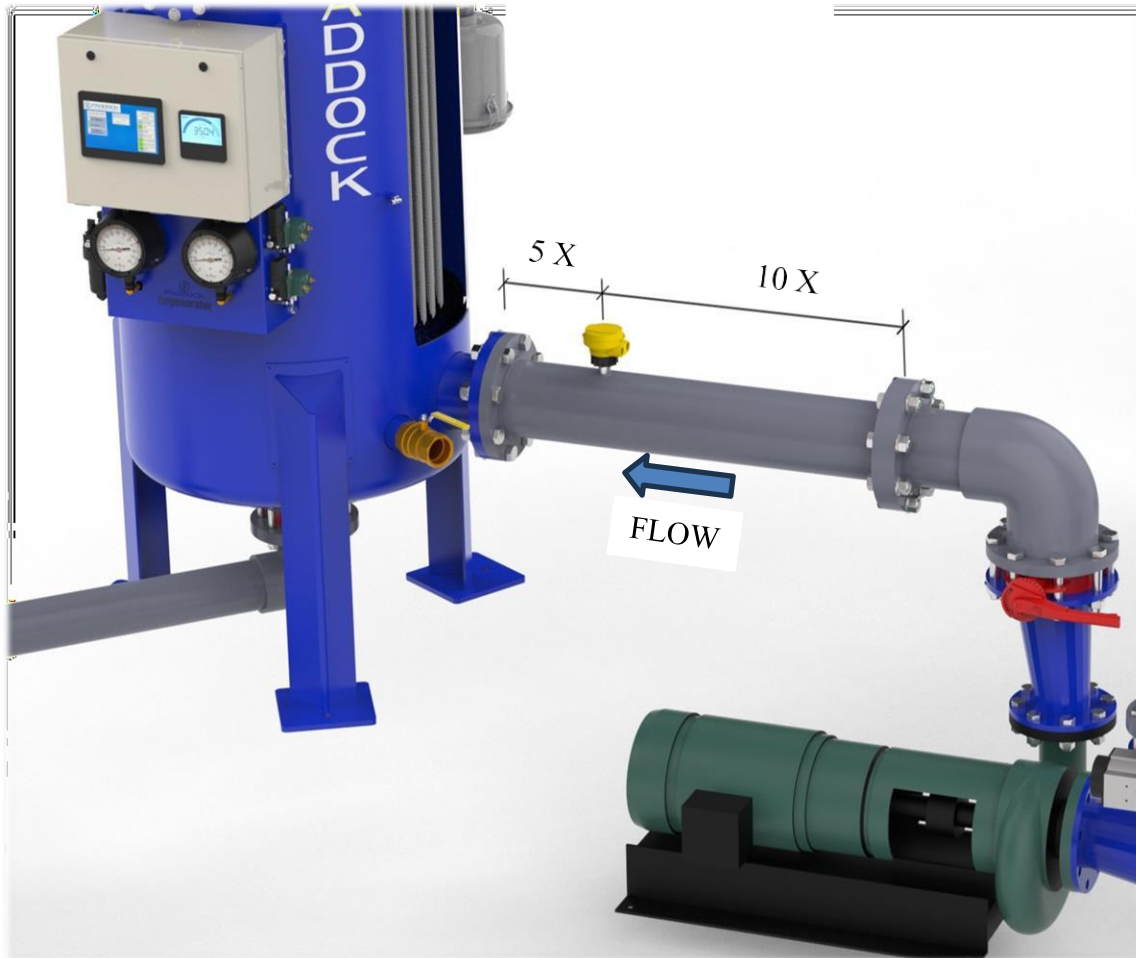


INCORRECT VALVE LOCATION



Flow Meter Location

The flow meter should be installed on the straight run of pipe into the influent connection between the pump and the filter. This will allow the flow to be monitored through both the pre-coat and on-stream cycles. The flow meter requires a distance of 10x the diameter of the pipe before and 5x the diameter of the pipe after.



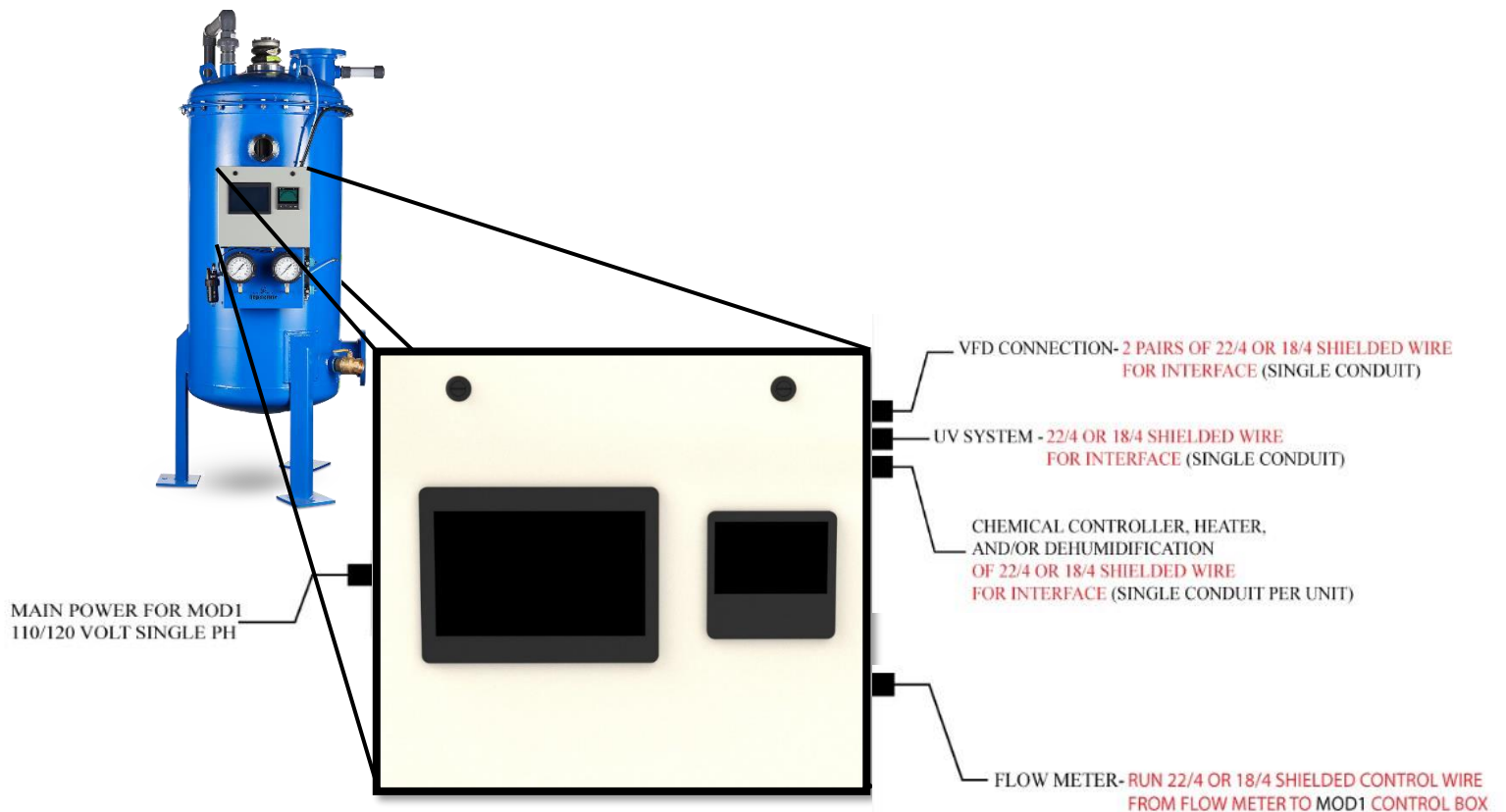
Electrical Requirements

The control panel requires a dedicated 120V single phase 20-amp circuit for power.

Control wiring requirements:

- 3 pair shielded cable from control panel to VFD
- 2 pair shielded cable from control panel to flow meter
- 1 pair shielded cable from the control panel to auxiliary systems requiring an interlock (heater, UV, etc.).

All electrical penetrations should be made on the sides or bottom of the controller. The filter and all other equipment need to be bonded.



Air Supply

The Paddock PPEC line of Regenerative filters utilize two pneumatic valves and a pneumatic “bump” mechanism. For operation, the filter requires a continuous supply of dry air at 80 PSI.

Air Compressor

A 60-gallon air compressor should be used to supply the filter with air for proper operation. For single filter installations, models PPEC 145S-900S will only need the compressor. For models PPEC 1275-2100 a 30-gallon receiver tank will be required. For installations with multiple filters, a receiver tank or even a second compressor may be required. The air supply from the compressor to the filter needs to be 3/4” braided 300 PSI hose or 3/4” copper. If using copper, do not heat fittings within two feet of the threaded connection to the filter. Direct heat can cause damage to the regulator components.

Air Dryer

An Air Dryer should be installed and plumbed between the air supply and the filter. This will help pull any remaining moisture out before it reaches the filter.



Air compressor

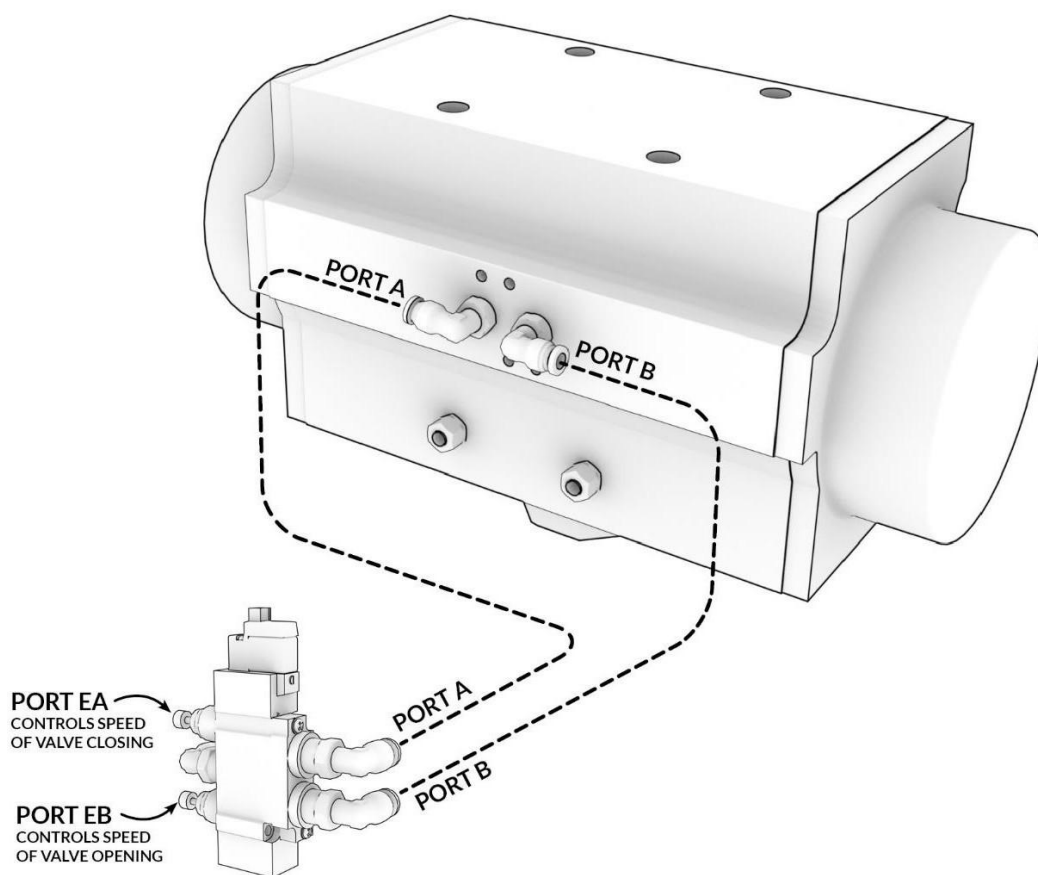
Receiver Tank

Air Dryer

Air Lines to Valves

The pneumatic valves should be connected to their respective control solenoid located on the right side of the filter control mounting bracket (top is On-stream, bottom is Pre-coat).

Speed control set screws for the valves are located on the front of the solenoid and should be adjusted for 3-5 seconds open and closure rate.





Operation & Maintenance Manual



“PPEC” **Regenerative** Series Filter

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FILTER OVERVIEW

“PPEC” Regenerative Series Filters

SCOPE

The filter in this manual is a Regenerative pressure diatomite or perlite design, manufactured by Paddock Pool Equipment Company, Inc. It is capable of sustaining the application now throughout the entire filter cycle without continuous body feed.

Each filter is a standard product of Paddock Pool Equipment Company, manufacturer of filtration equipment. Primary filter components include a tank, Flexible filter element assembly, and Regenerator™ "Bump" mechanism. All components and related subassemblies are factory assembled and tested prior to shipment.

PRINCIPLE OF OPERATION

Liquid travels through the filter in an up-flow direction. Unfiltered liquid typically enters a side connection in the tank, travels through a diffuser, then upward through the flexible filter element assembly leaving the filter via an outlet in the head.

When the system is initially started, the filter is pre-coated with diatomite or perlite filter media introduced through a vacuum transfer system. Pre-coat is pumped through filter where media is retained on the outer surface of elements forming the filter's "working surface." Similarly, particulates in the flow stream are held on the surface of the filter media while liquid passes through.

The filtration process is a mechanical straining of liquid through a porous layer of filter aid.

After a period of operation, accumulated solids gradually lessen the porosity of the "working surface" resulting in a decrease in effluent flow.

By activating the "Bump" mechanism, flow is restored and filter cycle is readily extended without backwashing or filter media change. This process, termed regeneration, repositions solids-laden pre-coat within the filter cake while randomly exposing new filter surface so the flow of liquid is no longer impeded. Following regeneration, filtration continues at full flow and at approximately original influent pressure. All "PPEC" series filters are capable of extended cycle operation and complete utilization of pre-coat. These automated filters provide programmed control of the regeneration cycle.

When solids-holding capacity of filter is finally reached (determined when regeneration no longer restores flow and pressure), the particulates and spent filter media are bumped from filter elements and drained from the tank. Reverse flow backwashing is neither necessary or recommended.

FILTER TANK

Filter tank is diametrically divided into head and body components. The two are bolted together and made watertight by an O-ring seal.

Both body and head are of welded construction, with all wetted surfaces and connections fabricated from low carbon stainless steel. External brackets and support legs are also fabricated from stainless steel.

Tanks are designed for a 50 PSI working pressure using a safety factor of 4 to 1. All tanks are hydrostatically tested to 1.5 times the design working pressure.

Standard tank construction incorporates connections for filter influent, effluent and drain, pre-coat inlet, pressure and vacuum relief, instrumentation, and inspection window and lift shaft gland.

Inspection window is made of Pyrex glass, and is covered by a clear plastic safety shield.

ELEMENT ASSEMBLY

Principal components include a tube sheet, flexible filter elements, seals and a lift shaft. These combine to diametrically separate filter tank into upper and lower sections. Seals prevent unfiltered water from bypassing to the upper section, clean side of the filter. All components in assembly are constructed from materials inherently passive to the application.

For servicing, assembly can be removed from filter tank as a unit, or, if desired, by dismantling individual parts.

In its operating format, the entire assembly moves down and up on a vertical axis. During filtration it is in an up or sealed position so unfiltered liquid must pass through flexible filter elements before it can leave the tank. When bumping, the entire assembly slowly moves to a down position, and then is rapidly raised.

FLEXIBLE FILTER ELEMENTS

Flexible filter elements are the porous support structures for diatomite or perlite filter media. Cylindrical in shape, each element is closed at bottom and open at top. The open end of a filter element is flanged and used for attaching each element to the tube sheet.

Outer wall of the filter element is made of filaments arranged so that external pressure causes a diminution of the tube diameter and pore size. Conversely, internal pressure results in an enlargement of diameter and pore size. Each element has an internal stainless steel spring to limit diameter diminution. During filtration the filter element is retentive. When bumped for regeneration or cleaning, it freely responds to internal pressure generated by a “Bump” stroke, propelling filter media and accumulated solids from its surface.

Filter elements have a maximum recommended operating differential of 20 psi and an ultimate of 75 psi.

“BUMP” MECHANISM

The “Bump” mechanism includes a pneumatically actuated Air-Stroke™ mounted on the head of the filter. It is connected to the flexible element assembly by a stainless steel shaft.

During bumping the Air-Stroke™ is alternately de-pressurized, then pressurized causing the flexible element assembly to move in a downward, then upward fashion. Total stroke travel is about 1-1/2 inches.

Bumping is operator-initiated on manual systems, and controller initiated on automated systems.

MODULAR SYSTEMS

System building through modular application of two or more filters is recommended. Dedicated pumps are used for each filter module. Pumps are sized for an equal portion of the total system flow rate. Thus, each filter module functions independently of each other.

APPLICATION DATA

MODEL	TANK DIAMETER	WORKING PRESSURE	TANK MATERIAL	INLET OUTLET	DRAIN
PPEC 145	18 IN.	50 PSI	304L	4X9	4X9
PPEC 225S	24 IN.	50 PSI	304L	4X9	4X10
PPEC 350S	30 IN.	50 PSI	304L	6X11	4X11
PPEC 500S	36 IN.	50 PSI	304L	6X11	4X12
PPEC 700S	42 IN.	50 PSI	304L	8X13.5	4X13
PPEC 900S	46 IN.	50 PSI	304L	8X13.5	4X14
PPEC 1275	48 IN.	50 PSI	304L	8X13.5	4X15
PPEC 1400S	56 IN.	50 PSI	304L	10X16	4X16
PPEC 1675	60 IN.	50 PSI	304L	10X16	4X17
PPEC 2100	60 IN.	50 PSI	304L	10X16	4X18

Standard tank construction is non-code, safety factor = 4

OPERATING DATA

MODEL	GPM RANGE	PRECOAT LBS PERLITE	PRE-COAT LBS DE	*TOTAL HEAD LOSS CLEAN	BUMP SYSTEM	OPER. WEIGHT
PPEC 145	49-212	12.5	24	2.5 PSI	Elec-Air	800
PPEC 225S	100-229	26	50	2.5 PSI	Elec-Air	1500
PPEC 350S	168-721	41	79	2.5 PSI	Elec-Air	3200
PPEC 500S	249-1066	65.5	126	2.5 PSI	Elec-Air	4250
PPEC 700S	339-1452	82	158	2.5 PSI	Elec-Air	5700
PPEC 900S	789-1691	95	190	2.5 PSI	Elec-Air	6000
PPEC 1275	896-1920	109	210	2.5 PSI	Elec-Air	7700
PPEC 1400S	549-2351	135	260	2.5 PSI	Elec-Air	9900
PPEC 1675	582-2496	141.5	272	2.5 PSI	Elec-Air	1200
PPEC 2100	737-3158	182	350	2.5 PSI	Elec-Air	13800

*Measured at gauge taps located on filter body and head

FLEXIBLE FILTER ELEMENT DATA



The flexible element is a porous support structure for diatomite or perlite filter aid used in Paddock Pool Equipment Company Regenerator™ filters. Several elements are vertically installed within the filter to form a tube nest assembly.

The outer wall of an element is made of braided groups of filaments arranged at a specific angle. Thus, external pressure causes a diminishing of tube diameter and pore size. Each element has an internal stainless-steel spring to limit diameter diminution. Filament material varies with application.

During filtration the element is retentive. Conversely, when bumped for regeneration or cleaning the element responds to internal pressure generated by a “Bump” stroke, propelling filter media and accumulated solids from its surface.

Flexible elements have a recommended operating differential of 20 psi and an ultimate of 75 psi.

Model	FLEXIBLE ELEMENTS	BARE AREA	COATED AREA
PPEC 140	126	52.4	77.3
PPEC 145	168	69.9	103.2
PPEC 225	340	111.9	165.1
PPEC 225S	340	141.4	208.7
PPEC 350	572	188.3	277.7
PPEC 350S	572	237.9	351.2
PPEC 500	846	278.4	410.7
PPEC 500S	846	351.9	519.4
PPEC 700	1152	379.2	559.3
PPEC 700S	1152	479.2	707.3
PPEC 900	1342	441.7	651.5
PPEC 900S	1342	558.3	819.0
PPEC 1000	1524	501.6	739.9
PPEC 1275	1524	633.9	935.8
PPEC 1400	1871	616.1	908.8
PPEC 1400S	1871	778.8	1141.0
PPEC 1675	2506	824.8	1216.6
PPEC 2100	2506	1042.5	1538.8

REGENERATOR™ "BUMP" MECHANISM

DESCRIPTION

Paddock Pool Equipment Company's Regenerator™ filters utilize a DOWN-UP stroke of the flexible filter element assembly to cause a relative flow reversal through filter elements. This process, termed "Bump" cycle, enables periodic reorganization of filter media for maximum utilization of the media and extended filter cycles. It also assures resource efficient cleaning of the filter without resorting to conventional reverse-flow backwashing.

To transmit force needed for "Bump" stroke, internal filter parts are connected to an Air-Stroke™ actuator mounted on top of filter head. Stops are provided in both directions in order to achieve a specific stroke length.



OPERATION

Downstroke (compression stroke) allows gravity (weight of parts) to lower element assembly. Upstroke (extension stroke) is accomplished pneumatically by quickly filling actuator with compressed air.

Since actuator and element assemblies are mechanically connected, extension of Air-Stroke™ causes a rapid lifting of the element assembly to its upstroke position. This movement of the element cluster through an incompressible surrounding liquid propels filter media and accumulated particulates from the elements.

Regenerator™ "Bump" cycling is automatically initiated by the filter controller, while cleaning cycles are operator-initiated.

MECHANICAL

While Air-Stroke™ actuator can operate without damage through an arc of up to 30"; its application in Paddock Pool Equipment Company's "Bump" mechanism is along a single vertical axis, thus assuring a long service life. Compressed air used to fill and exhaust actuator is controlled by a 3-way solenoid valve triggered by control circuits in filter controller.

OPERATING LIMITS

- Pressure: 90 psi maximum
- Temperature: -35 F to +135 F
- Electrical Requirements: 1-60-120 VAC, 6.8 Watts
- Service Life: Exceeds 10 million cycles
- Maintenance: Maintain in an externally clean condition
- Air-Stroke™ Adjustment: None
- Compressed Air Requirements: 2 scfm @ 90 psig; 1" to 1-1/2" supply to regulator depending on length of pipe run

CAUTION

Never inflate actuator when it is unrestricted; i.e., with compression and extension stops removed.

Do not inflate beyond specified pressure limit.

Improper use or over-inflation may cause a failure of the actuator with component damage or resulting in personal injury.

PPEC 225-350 AIR FILTER/ REGULATOR

Ref. Parker Hannifin "Piggyback"

Features:

- A. Space-saving "Piggyback" combines all filter and regulator features to yield optimal performance
- B. High water-removal efficiency
- C. Quick response and accurate pressure regulation regardless of changing flow or inlet pressure

OPERATION:

If no air is supplied to unit, piston and valve poppet are held in open position by regulator's adjusting spring. Pressurized air flows through louvered deflector and is directed into a swirling pattern. Liquids and large dirt particles are thrown against inside wall of bowl and fall into "quiet zone" below lower baffle. The baffle prevents liquids and solids from re-entering air stream. Clean air passes through element and open poppet valve to outlet port. Pressure increases in outlet line and below regulating piston, causing piston to move against the adjusting spring. The poppet, following this movement, closes and stops air flow as soon as preset pressure level is reached. This pressure level is determined by adjusting spring through T-handle. Pressure level is now controlled automatically. Valve opens for more flow if secondary pressure decreases. Excess pressure causes regulating piston to move away from relief tube. Air can exhaust through vent hole in the piston and bonnet. (This happens for standard relieving type only.)



AIR LUBRICATOR

Ref. Parker Hannifin Mist Lubricate

Features:

- A. Lubricates through broad air flow range
- B. Venturi By-Pass disc develops a linear oil delivery as air flow varies
- C. Precision needle valve results in repeatable oil delivery and minimum drip-rate change
- D. Bowl can be filled while air-line is under pressure
- E. Reverse flow capability

OPERATION:

Some of the air entering lubricator is passing through a nozzle at increased velocity. This creates a pressure differential which is used to inject oil into air stream. Higher pressure in the bowl forces oil through a pickup tube, adjustment needle, and sight dome into the low-pressure area above nozzle.

The By-Pass disc deflects at higher flow rates and governs amount of injected oil proportionally to the air flow. Rate of lubrication can be adjusted by the precision needle valve. Rotation of the needle counter-clockwise will increase drip rate. Drip rate can be observed in sight dome. After adjustment, drip rate is controlled automatically with changes in air flow. The bowl can be filled through the fill cap, while lubricator remains pressurized.

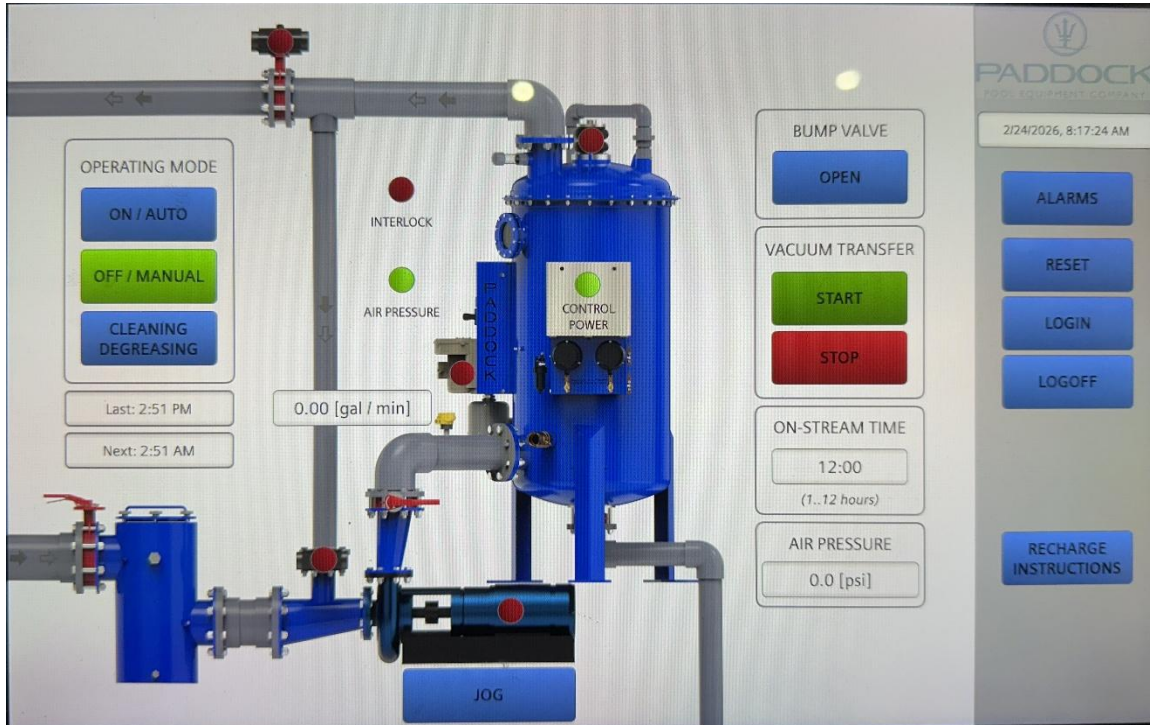


OPERATING INSTRUCTIONS

Filter Touchscreen

All PPEC Regenerative Media Filters are equipped with a 12" colored touchscreen for ease of and use. The below images provide an overview of the controller operation and setup.

Home Screen



OPERATING MODES

- **ON/AUTO** – This mode turns the filtration system on and automatically controls its operations.
- **OFF/MANUAL MODE** – This mode turns the filtration system off and used during recharging of the filter and to adjust the On-stream time. When selected, the motor will stop, On-stream valve will close, and interlocks will turn off. Also used to exit the CLEANING/DEGREASING mode.
- **CLEANING/DEGREASING** – This mode is for degreasing and demineralizing the filter. When placed in this mode, the filter will continuously bump at a predetermined interval until the OFF/MANUAL mode is selected.

BUMP VALVE

This is a momentary button is used to manually bump the filter during cleaning cycles. When button is pressed and held, the bump mechanism will drop. When button is released, the bump mechanism will rapidly stroke up.

VACUUM TRANSFER

Start and stop buttons control vacuum transfer motor used during media recharge, degreasing, and demineralizing processes.

ON-STREAM TIME

Selects the number of hours between bump cycles. Pressing the number will provide a numerical keypad to set the time between 1-12 hours.

AIR PRESSURE

Displays current air pressure filter regulator is set to. When pressure falls below 55 Psi an alarm will trigger.

PUMP JOG

This momentary button is used to manually jog the pump when filling the filter with water.

MENU BUTTONS

- **ALARMS** – When selected will open alarm page to show active and cleared alarms.
- **RESET** – Quick reset of alarms without navigating to alarm page.
- **LOGIN** – Opens Username/Password dialog box for programming.
- **LOGOFF** – Logs out of programming.
- **RECHARGE INSTRUCTIONS** – Opens dialog boxes to help navigate through filter recharging procedure.

OPERATING DATA

Pre-Operating Checklist

- Power switches in panel – ON
- Enclosure door – SECURE
- Operating mode – OFF/MANUAL
- Vacuum Transfer – OFF
- On-Stream duration – SET
- Air supply to filter – ON
- Regulator set to proper setting – 90 PSI

CAUTION - TO PREVENT ELECTRIC SHOCK

DISCONNECT SUPPLY CIRCUIT(S) BEFORE ATTEMPTING SERVICE TO THE SYSTEM OR CONTROLLER; AND BEFORE OPENING ENCLOSURE DOOR

IMPORTANT INFORMATION

1. Automatic Program can be interrupted at any time. On repowering, the controller will automatically initiate a new filter cycle beginning with a “BUMP” cycle.
2. The PPEC filter controller incorporates a system air pressure sensing circuit. If the system air pressure falls below set point, and alarm will trip, the screen will turn red, and the controller will STOP filter cycle. Upon restoration of normal air pressure, controller will restart filter beginning with a “BUMP” cycle.
3. If power is interrupted to the filter controller, once it is restored, it will revert to the OFF/Manual mode. The operator will need to place it back into ON/AUTO mode
4. If a fault occurs on the filter controller while it is in its ON/AUTO mode, once the fault has been corrected, it will automatically restart beginning with a “BUMP” cycle.

MAINTENANCE & CARE

FILTER MAINTENANCE

ESTABLISH A SCHEDULE

Once the filter reaches 10-12 PSI differential after bumping, the solids-holding capacity of filter is finally reached and spent filter media should be bumped from filter elements and drained from the tank. The chart below shows quantity of media needed per each model filter.

MODEL	PRECOAT LBS PERLITE	PRECOAT LBS DE
PPEC 145	12.5	24
PPEC 225S	26.0	50
PPEC 350S	41.0	79
PPEC 500S	65.5	126
PPEC 700S	82.0	158
PPEC 900S	95.0	190
PPEC 1275	109.0	210
PPEC 1400S	135.0	260
PPEC 1675	141.5	272
PPEC 2100	182.0	350

DUMPING THE SPENT CHARGE

1. Manually turn off heaters and chemical control systems 5-10 minutes before shutting filter down
2. Push the OFF/MANUAL MODE button on filter controller. The box to the left should turn green.
3. Close pump discharge valve if below grade
4. Push and hold the green jog button for the Bump solenoid on the filter controller until actuator boot on top of filter is fully relieved of air. Release the button and repeat two more times
5. Open high air vent line valve
6. Open dump valve on bottom of filter and drain completely
7. Push the green jog button for the VFD on the filter controller for 5-8 seconds to jog the pump washing out any solids left in bottom of filter. If below grade, opening and closing pump discharge valve for the same duration should be sufficient.

8. Close dump valve on bottom of filter and use the VFD jog button or open pump discharge valve if below grade to slowly refill filter completely (water should come out of the air vent line). Close air vent line valve and repeat steps 3-7 one more time
9. After completing step 7 for the second time, close dump valve, close air vent line, and open Vacuum Transfer drain valve letting the line drain completely. (Illustration of Vacuum Transfer valve orientations found on following page of this manual) If below grade, pump discharge valve should still be closed.

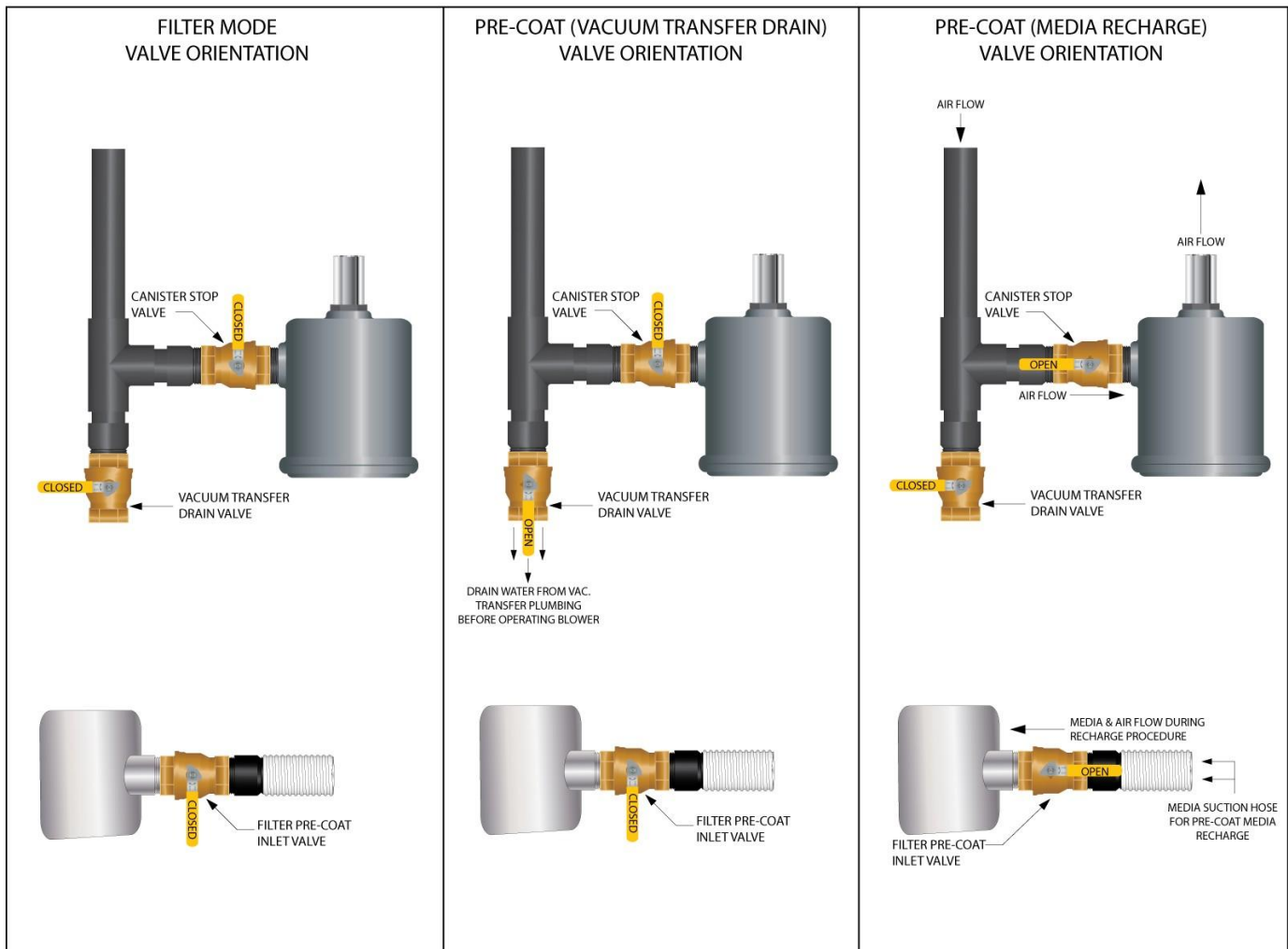
RECHARGING FILTER

1. Close Vacuum Transfer drain valve and open Canister Stop valve and Filter Pre-Coat Inlet valve. Make sure the Vacuum Canister lid is securely fastened (illustration of valve orientation found on following page of this manual)
2. Press the green start button for the Vacuum Transfer on the filter panel and vacuum in proper quantity of media. Close Canister Stop and Filter Pre-Coat Inlet valves. Hang Vacuum Canister lid to allow any moisture to dry.
3. Open air vent line valve and use the VFD jog button or open pump discharge valve if below grade to slowly fill filter until a steady stream of water comes out of air vent line.
4. Close air vent line
5. Push the ON/AUTO button on filter controller. The box to the left should turn green and the sequence start
6. Monitor filter as it goes through pre-coat cycle and return to pool
7. Turn on heaters and chemical control systems

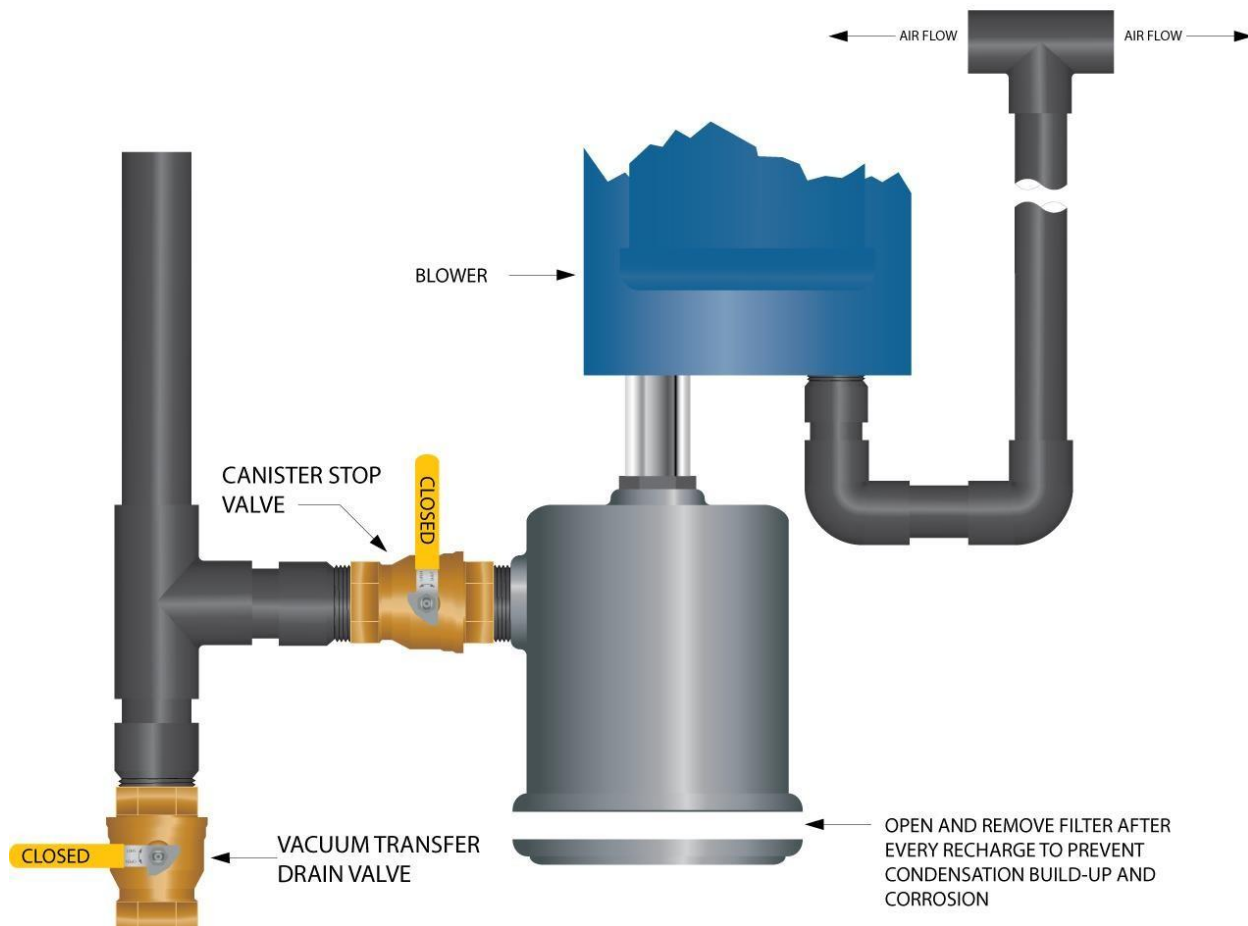
VACUUM TRANSFER SYSTEM

The Vacuum Transfer system is used to introduce media and cleaning agents into the filter. The following diagrams display the valve orientations on the Vacuum Transfer piping for proper operation.

VACUUM TRANSFER VALVE ORIENTATION

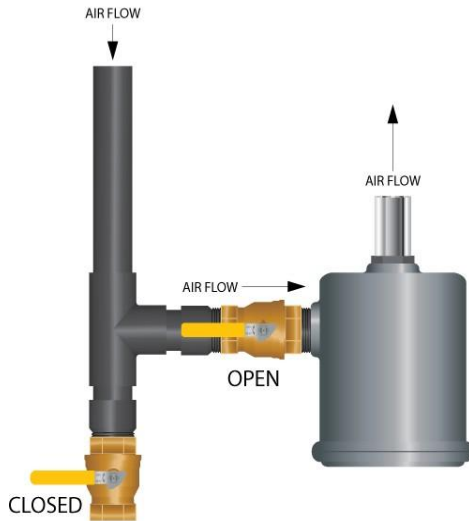


VACUUM TRANSFER FILTER CLEANING AND OPERATION



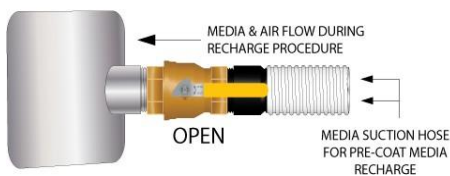
VACUUM TRANSFER INSTRUCTIONS

PRE-COAT (MEDIA RECHARGE) VALVE ORIENTATION



RECHARGING FILTER IS AN EASY PROCESS AND ONLY REQUIRES A FEW MINUTES. MAKE SURE SYSTEM IS OFF, ISOLATED, DRAINED AND RINSED PRIOR TO FOLLOWING STEPS.

1. DRAIN WATER FROM VACUUM TRANSFER PLUMBING (SEE VACUUM TRANSFER VALVE ORIENTATION DRAWING).
2. MAKE SURE HE PA FILTER IS IN FILTER CANISTER AND CLOSED
3. OPEN VALVES AS SHOWN
4. TURN VACUUM TRANSFER BLOWER ON (LOCATED ON FILTER CONTROLLER PANEL)
5. OPEN MEDIA BAGS AND BEGIN TRANSFERRING CORRECT AMOUNT OF MEDIA INTO FILTER BY USING VACUUM TRANSFER HOSE LOCATED ON FRONT OF FILTER. THE CORRECT AMOUNT OF MEDIA IS LOCATED ON FILTER LABEL AND IN OPERATORS MANUAL
6. ONCE ALL MEDIA HAS BEEN TRANSFERRED, TURN OFF VACUUM TRANSFER BLOWER, CLOSE ALL VALVES ON VACUUM TRANSFER AND FOLLOW VACUUM TRANSFER VALVE ORIENTATION DRAWING.



DEGREASING ELEMENT ASSEMBLY

While Paddock Pool Equipment Company Regenerator™ filters are inherently resistant to effects of contaminants normally found in swimming pools, it is important to remember that body oils, sunscreen lotions, and cosmetics tend to form gummy residues inside filters. In addition, mineral content in a pool increases daily as a result of continuous chemical treatment and normal water evaporation process. Accordingly, if contaminant levels are allowed to get too high, deposits can form on filter elements that may eventually shorten filter cycles.

To guard against this, degreasing and chemical cleaning of the element assembly is suggested at approximately **six-month intervals**.

The most effective means for removing gummy residues of body oils, sunscreen lotions, and cosmetics is through a soak cycle using a cold-water saponification, such as Paddock Pool Equipment Company Sodium Percarbonate (Degreasing Concentrate).

To initiate procedure:

- Perform normal filter media dump cycle and rinse as described previously.
- Using the Vacuum Transfer system, add prescribed amount of Degreasing Concentrate (see table next page), then fill the filter tank slowly utilizing the green “Jog VFD” button.
- Press the “Cleaning/Degreasing” button (box to the left should turn green).
- Allow solution to remain in contact with elements for a minimum of 8 hours while in “Cleaning/Degreasing” mode.
- **High vent bleed valve must remain fully open during the degreasing cycle. Failure to do so will result in pressurization of the tank causing tank failure and/or injury!**

After completing soak cycle, drain solution from filter tank. Refill and drain filter at least twice with pool water to thoroughly flush all wetted parts.

SPEED CLEANING PROCEDURE

Following the procedure outlined in degreasing the element assembly:

1. Place filter in ON/AUTO mode
2. Let the filter bump and then precoat for 10-15 seconds
3. Press the OFF/MANUAL button, then the CLEANING/DEGREASING button
4. Repeat steps 1-3 at hourly intervals for 4 hours
5. Drain and rinse (2) times
6. Charge filter and operate normally

CLEANING / REMOVING MINERAL DEPOSITS

To complete maintenance procedure, a chemical cleaner such as Paddock Pool Equipment Company Citric Acid (Demineralizer) Again, fill filter with pool water and allow solution to remain in contact with elements for a minimum of 8 hours in its "Cleaning/Degreasing" mode.

After completing soak cycle, drain solution from filter. Refill and drain filter at least twice with pool water to thoroughly flush all the wetted parts. Filter is now ready to go back into service.

GUIDELINE CHART QUANTITY

The following table should be used to determine amount of Degreasing Concentrate and Chemical Cleaner needed for each model.

Filter Model	Tank Volume (gals.)	Note 1	Note 2
		Degreasing Concentrate (lbs.)	Demineralizer (lbs.)
PPEC140	89	6.4	4.0
PPEC145	98	8.0	6.0
PPEC225S	129	17.6	10.0
PPEC350S	244	31.7	17.0
PPEC500S	291	41.6	24.0
PPEC700S	396	59.2	34.0
PPEC900S	496	64.5	35.0
PPEC1275	595	77.0	42.0
PPEC1400S	721	93.7	50.0
PPEC1675	801	126.4	75.0
PPEC2100	890	149.0	90.0

Note 1: Suggested pounds of Paddock Pool Equipment Company Degreasing Concentrate (packaged in 50-lb bag)

Note 2: Suggested pounds of Paddock Pool Equipment Company Demineralizer (packaged in 50-lb bag)

SOME PRECAUTIONS

Take extreme care when handling cleaning agents for they can be harmful to the eyes, skin and clothing. [Read manufacturer's label instructions and Safety Data Sheet \(SDS\) before using.](#)

Isolate filter from the pool and recirculation system by closing valves, etc. before introducing cleaning solutions to filter.

Caution:

The use of muriatic acid as a cleaning chemical should be **AVOIDED**. Muriatic acid, even in weak concentration, will quickly corrode most metals including stainless steel tank. It will also destroy filter elements.

For further information or question(s), see below to contact Paddock Pool Equipment Company, Customer Service.

PPEC REGENERATOR FILTER COMPONENTS

Included in Regenerator Pkg

Item	Name
R1	PPEC Regenerator Filter
R2	Pneumatic On-Stream Valve
R3	Pneumatic Precoat Valve
R4	Precoat Fill Valve
R5	Flow Meter
R6	Control Panel
R7	Bump Assembly
R8	Vacuum Transfer Pump
R9	Drain Valve

Optional Filter Accessories

F10	Check Valve
F11	Valve
F12	Strainer
F13	Reducing Precoat Tee
F14	Pump
F15	Reducer
F16	Air Compressor
F17	Auto Air Bleed
F18	Receiver Tank
F19	Air Dryer
F20	Air Line
F21	VFD
F22	Air Bleed

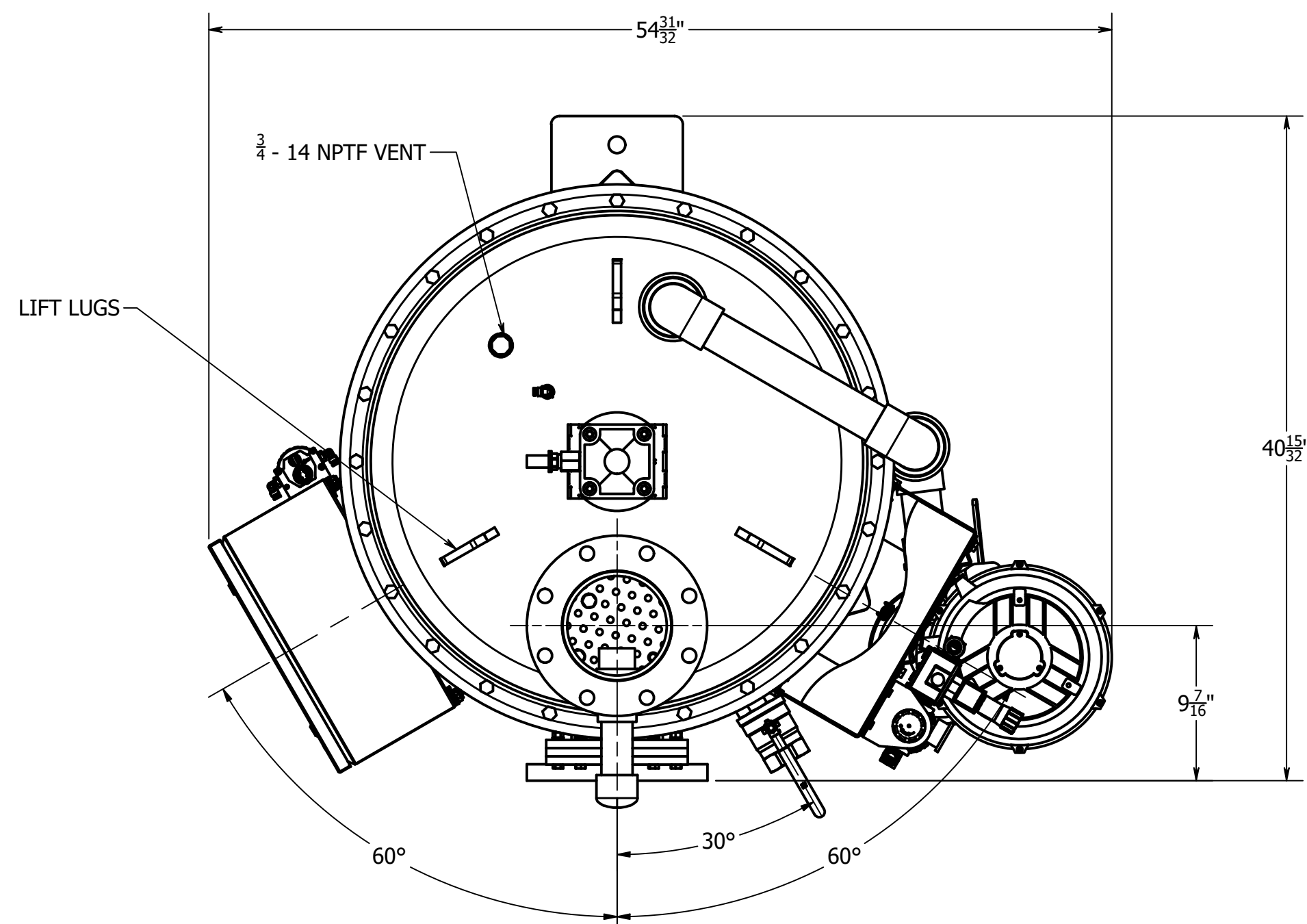


REGENERATIVE MEDIA FILTER SCHEMATIC

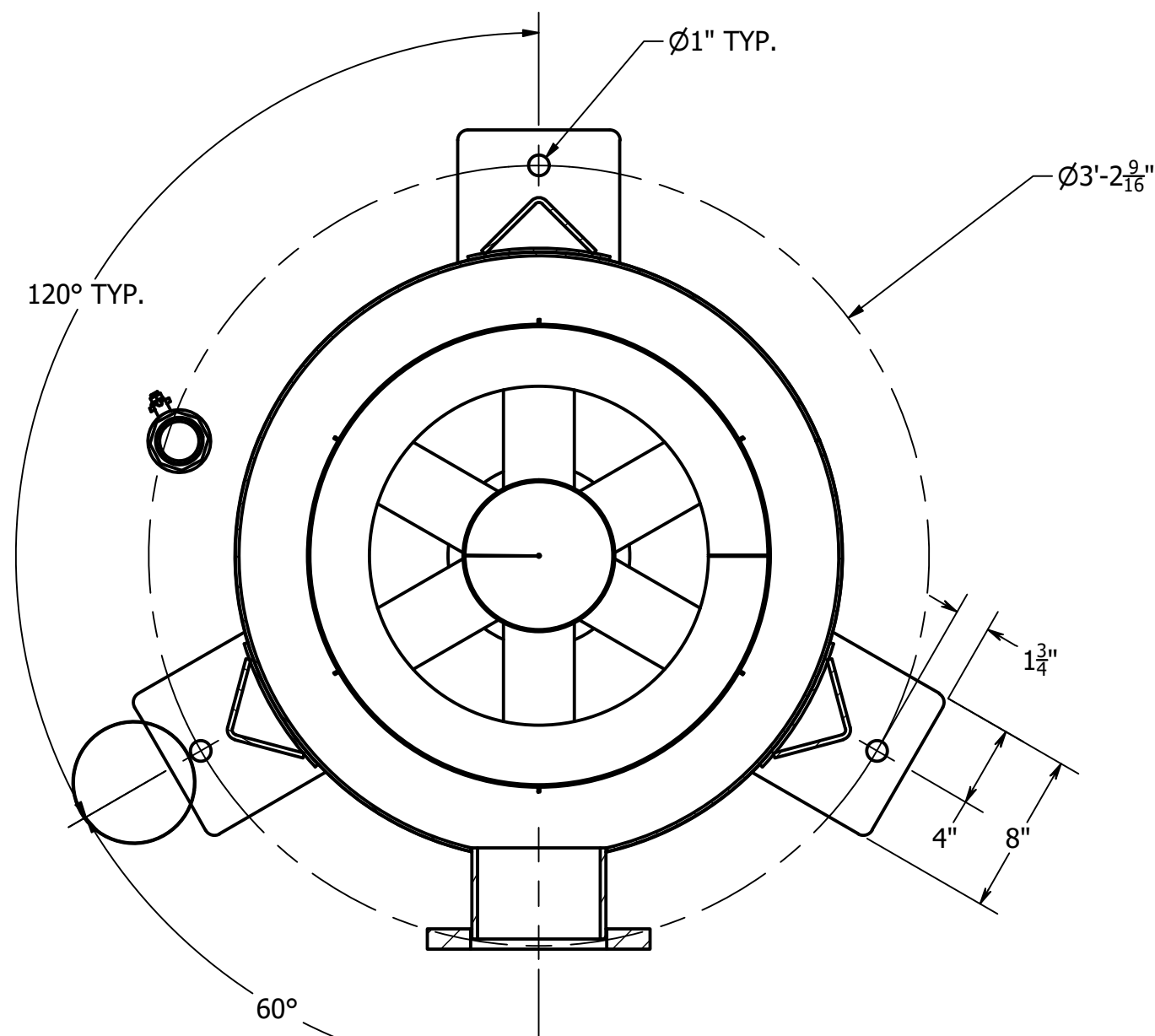
Filter Parts

1	Regulator – Specify Filter Model
2	Bump Solenoid
3	Vacuum Transfer Motor
4	Vacuum Canister
5	Brass Ball Valve
6	Bump Boot – Specify Filter Model
7	Filter Element
8	Viewport Window
9	Viewport O-Ring
10	Filter Controller
11	Lubricator
12	Pressure Gauge
13	Valve Solenoids
14	Drain Valve



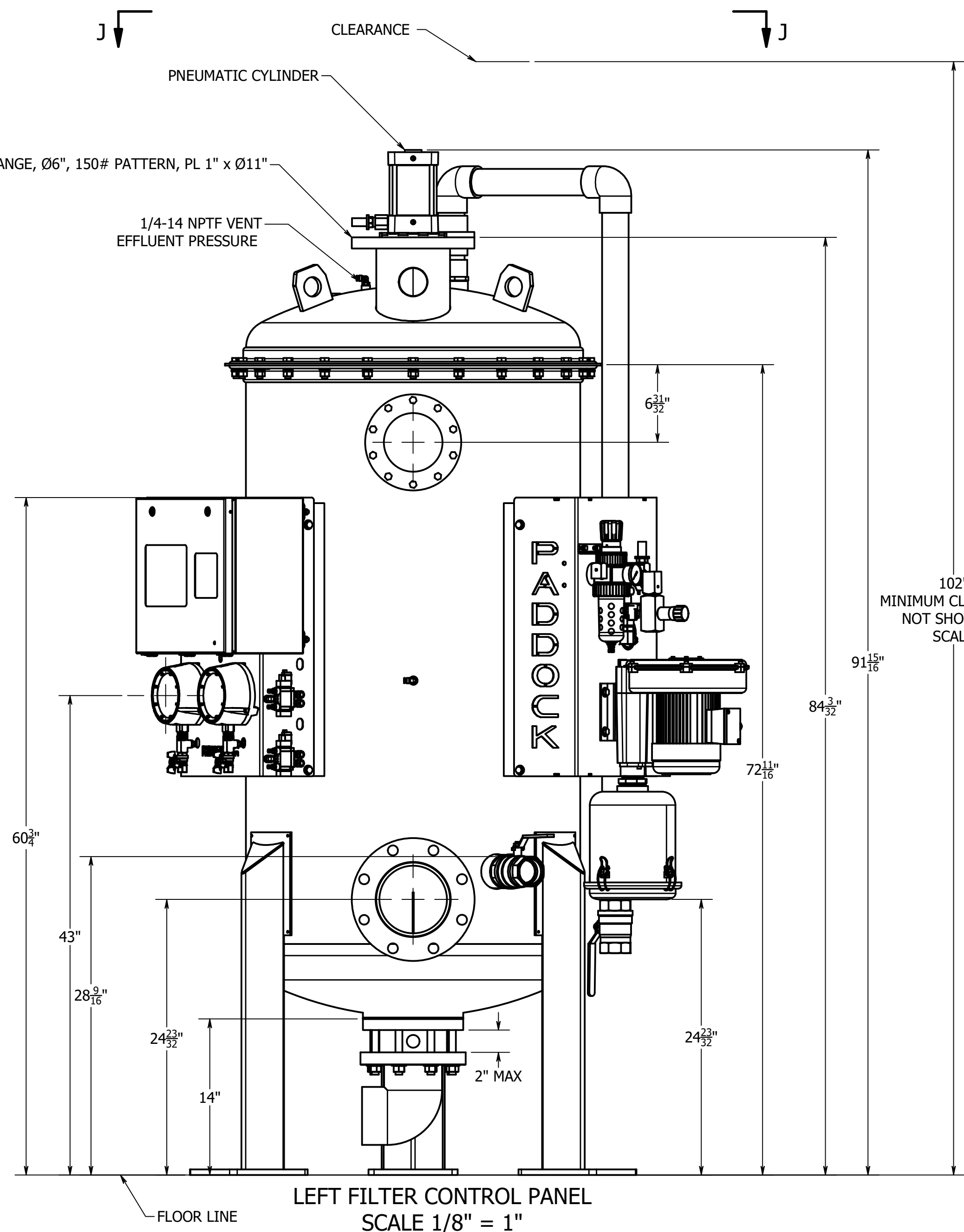
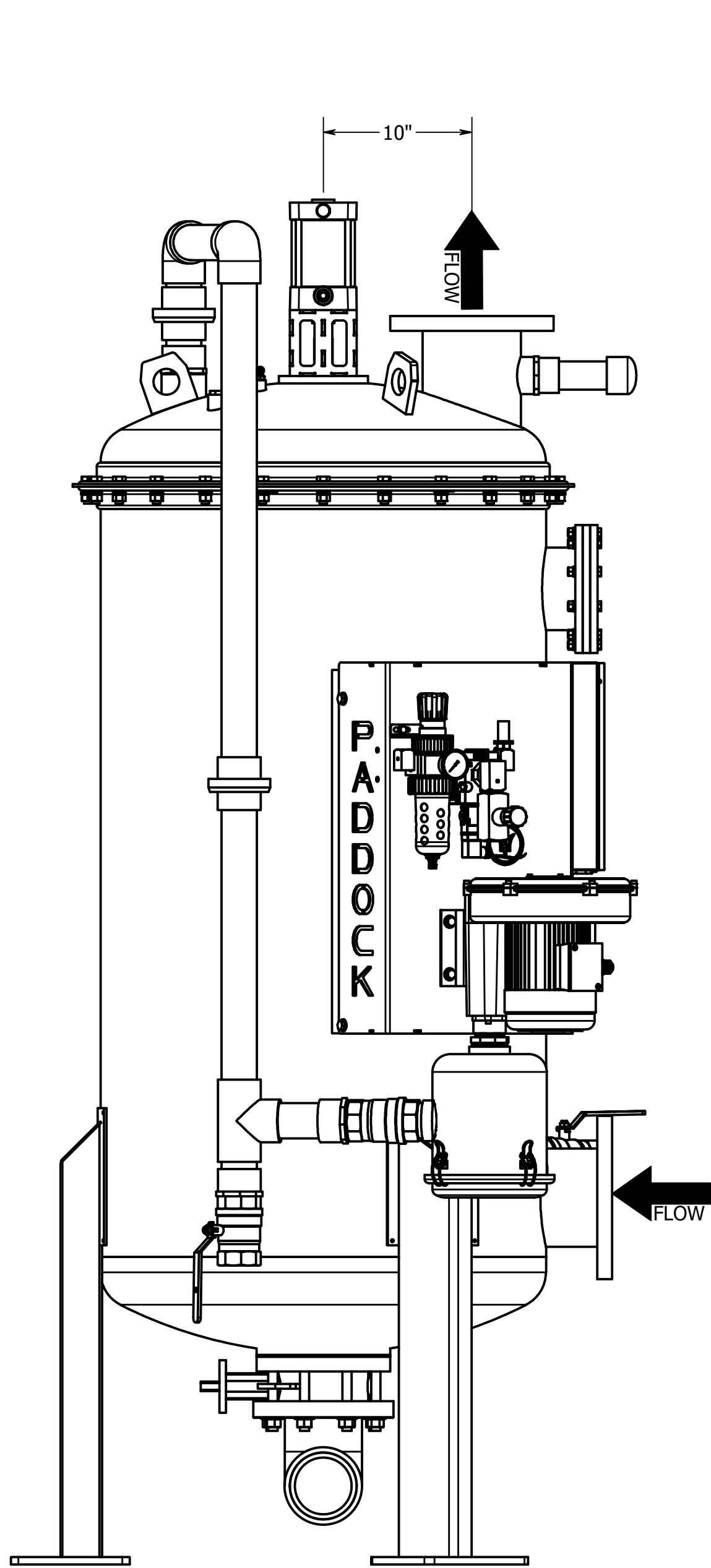


VIEW J-J
SCALE 1/8" = 1"

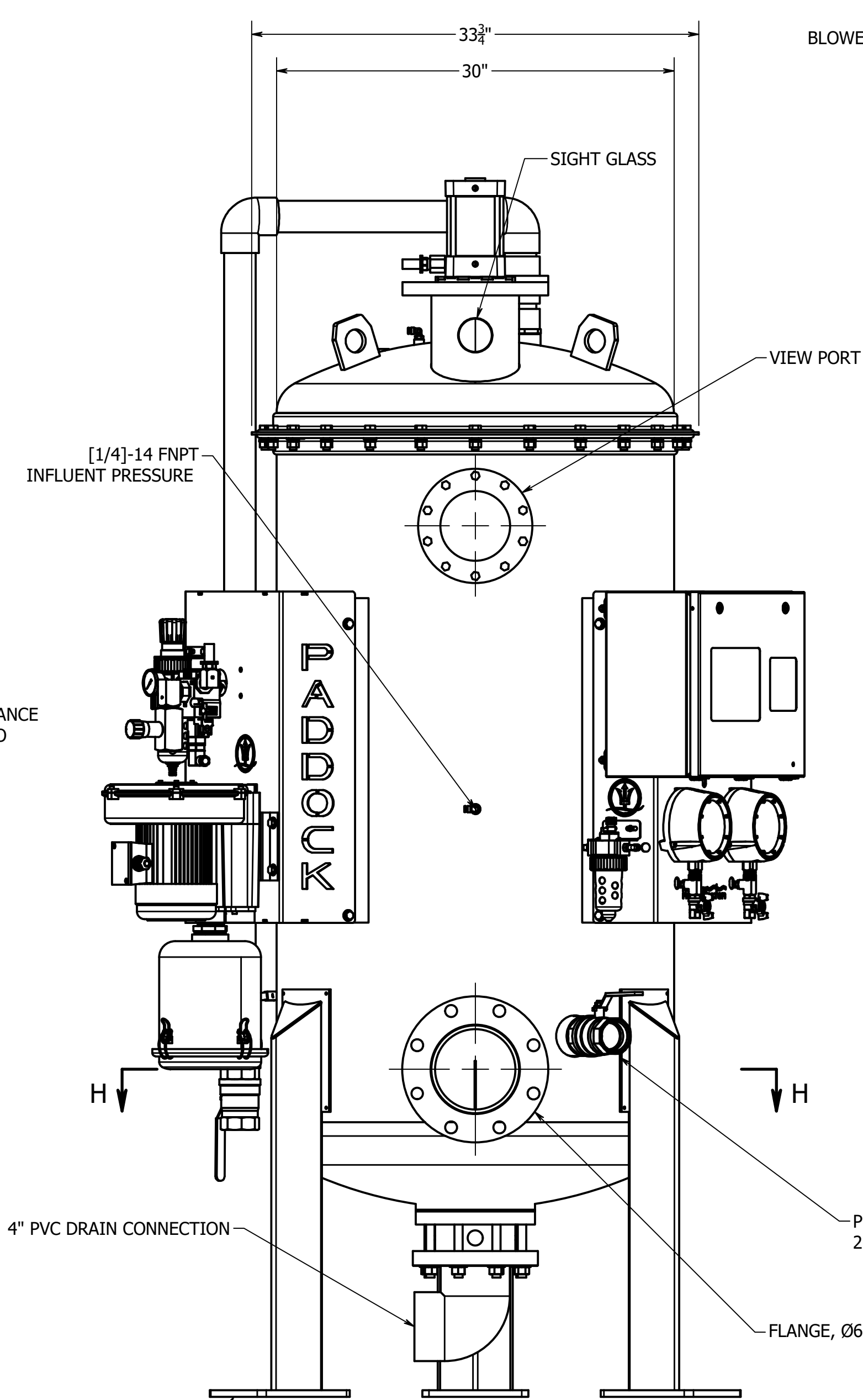


SECTION H-H
SCALE 1/8" = 1"

ASSEMBLY CONFIGURATIONS

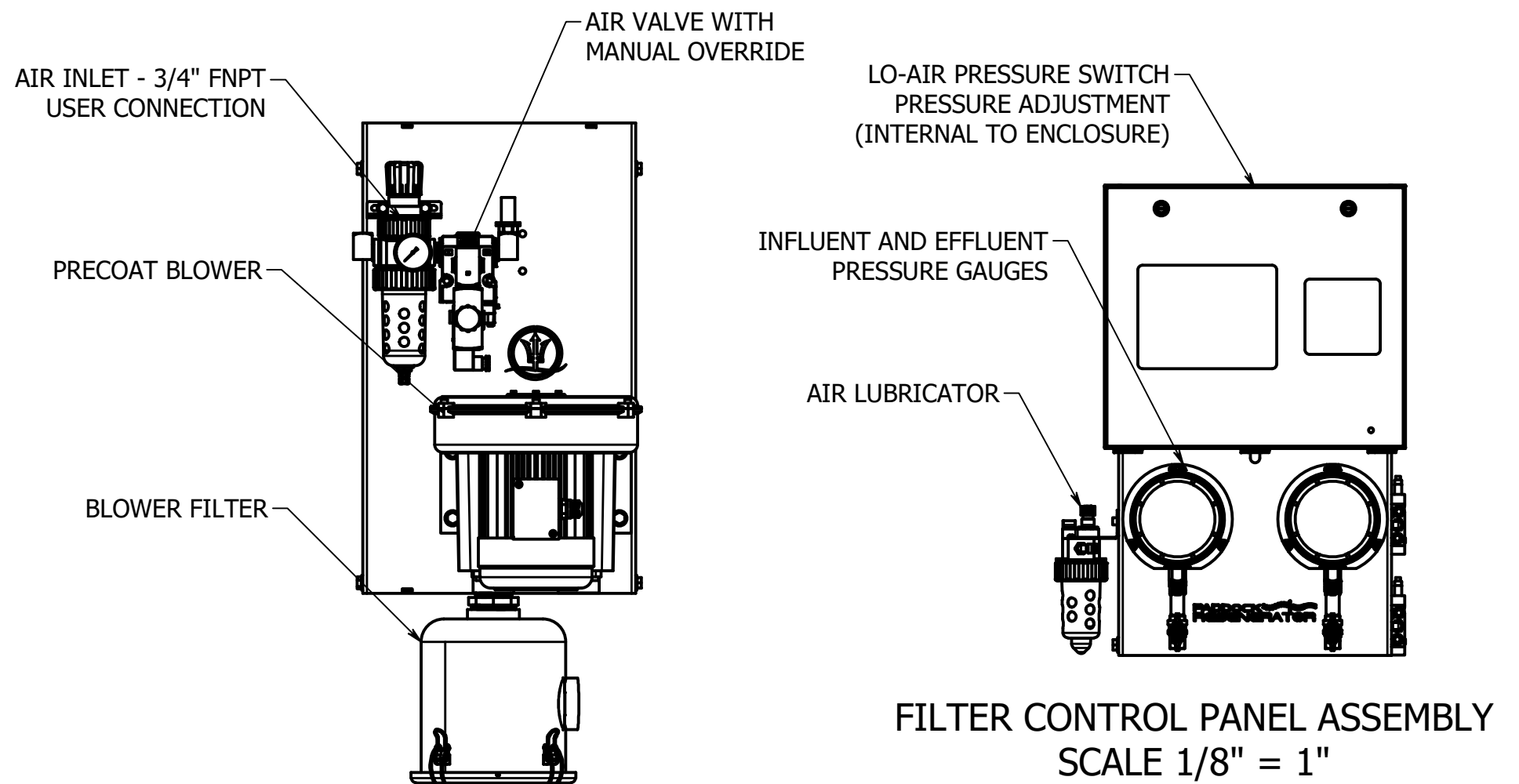


LEFT FILTER CONTROL PANEL
SCALE 1/8" = 1"



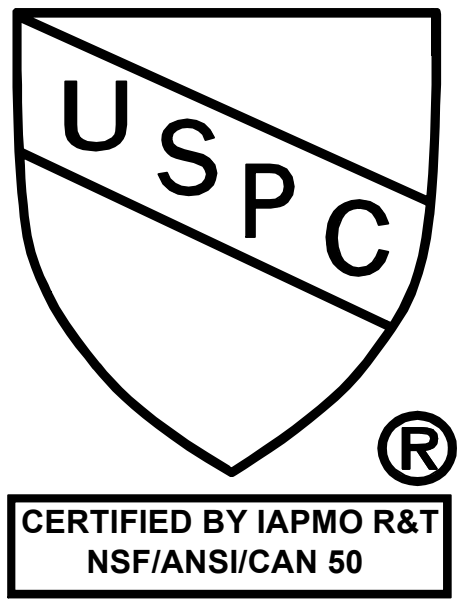
RIGHT FILTER CONTROL PANEL
SCALE 1/8" = 1"

ITEM		QTY	PART NUMBER	PARTS LIST	DESCRIPTION	COMMENTS
1	1	REG-SA0350S.01-R2	ASSEMBLY, REGEN, MODEL PPEC 350S	PPEC		
2	1	701325 with 6" Valve	ACTUATOR, WITH 6" VALVE	BRAY		
3	1	701209-024 with 4" Valve	ACTUATOR, WITH 4" VALVE	PPEC		
4	1	s30-4-p01	4" VALVE	BRAY		
5	1	s30-4-p02	4" VALVE	BRAY		
6	1	700455	NOTCH PLATE	304L SS		
7	1	700451	VALVE HANDLE	304L SS		
8	1	P2101-040x16.05-R0	FLANGE, HDPE 1" x Ø9"	HDPE		
9	1	700221	GASKET, DRAIN FLANGE	RUBBER		
10	1	700185-700598	ELBOW WITH FLANGE, SWIVEL 90	GPVC		
11	1	REGEN-0200VTH	2" VACUUM TRANSFER HOSE	PVC		
12	1	ADAPTER-0200MPTx0200BARB.08	ADAPTER, 2"MPT x 2" BARB	PVC		
13	4	SL-90-0008MPTx0008TUB.09-R0	ELBOW, 90°, 1/4" MPT x 1/4"	BRASS		
14	1	210024	ELECTRIC AUTO DRAIN	SPEEDAIRE		
15	1	701288	SENSOR, FLOW SIGNET2251, NO DISPLAY 5-8"	SIGNET		
16	1	701295	SADDLE, SIGNET 2551 6"	SIGNET		
17	1	201339	LIQUID BYPASS VALVE, 200PSI	HANKISON		
18	8	FW-063-316	FLAT WASHER, Ø5/8"	316 SS		
19	8	STUD-063Cx0508-316	STUD, 5/8"-11 X 5 1/2"	316 SS		
20	8	HNUT-063C-316	HEX NUT, 5/8"-11	316 SS		
21	3	200121	PERLITE, 50LBS BAG	PPEC		
22	1	200237	DEGREASER, 50LBS BAG	PPEC		
23	1	200236	DEMINERALIZER, 50LBS BAG	PPEC		



BLOWER PANEL ASSEMBLY
SCALE 1/8" = 1"

MODEL	MAX DESIGN FLOW	EFF.FILTRATION AREA/FT²	GPM RANGE
PPEC350	1.60	351.2	337-565



- NOTES:
- TANKS SHOULD NOT BE TURNED ON SIDE DURING TRANSPORT OR INSTALL.
 - ALL DIMENSIONS - INCHES ± 1/4"
 - ALL FLANGES - 125LB PATTERN

0	02/11/26	AG	RELEASED FOR FAB
REV	DATE	BY	DESCRIPTION
REVISION HISTORY			
DO NOT SCALE DRAWING			
TOLERANCE UNLESS OTHERWISE NOTED:			
X. ± 1/16"			
1/8" ± 1/32"			
X' ± 1/4"			
BY DATE			
DRAWN DTS 11/08/23			
CHECKED			
APPROVED			
DESCRIPTION			
POOL EQUIPMENT COMPANY			
555 Paddock Parkway Phone: (803)324-1111			
Rock Hill, SC 29730 info@paddockindustries.com			
REGEN, MODEL, PPEC 350S			
PPEC JOB NUMBER:			
DWG. NO. REG-0350S.304-R2			
SHEET # 1 OF 1			